Banner appropriate to article type will appear here in typeset article

Closure-channel identifiability and two-channel recovery in monatomic kinetic normal shocks

Ehsan Roohi[1]

[1]Department of Mechanical and Industrial Engineering, University of Massachusetts Amherst, Amherst, MA 01003, USA

Corresponding author: Ehsan Roohi, eroohi@umass.edu

(Received xx; revised xx; accepted xx)

Residual agreement in a heat-flux equation does not uniquely identify the higher-order moments entering a monatomic normal shock. We formulate this limitation as an observation-map problem: the heat-flux budget determines the composite fourth-order channel $S = R_{xx}^{\mathrm{cl}} + \Delta/3$, rather than the tensorial closure moment R_{xx}^{cl} and scalar fourth-order excess Δ separately. The map has a one-dimensional null space. A complementary scalar-excess budget, evaluated consistently with the discrete velocity method (DVM), supplies the missing information. For Bhatnagar–Gross–Krook (BGK) reference shocks at Mach 2, 3 and 5, two-budget recovery reduces the active-zone tensorial-moment error from approximately 63–64% to 2.54–4.43%. Exploratory moment evaluations at Mach 6, 8 and 12 retain a substantial scalar contribution. Sparse scalar measurements provide an information-reduction test whose accuracy depends on shock-core coverage. Shakhov, independently evolved finite-time ellipsoidal-statistical BGK (ES-BGK) shocks at Mach 2 and 3, ellipsoidal-statistical Fokker–Planck (ES-FP) source diagnostics, and direct simulation Monte Carlo (DSMC) diagnostics distinguish the collision-independent observation channel from collision-dependent scalar production. Completed Mach-5 DVM sensitivity tests give differences below 0.53% for the six tested fields. A separate DSMC sampler comparison at 960 particles per cell gives moment-profile differences of 0.73–0.93%, comparable to sampling uncertainty; the collision-source difference is 1.41%. The central result is an identifiability principle: recovering tensorial and scalar fourth-order closure information requires an independent scalar complement to the heat-flux observation channel.

8 ## 1. Introduction

A rarefied shock is a velocity-space object. Its density, velocity and temperature are low-order projections of the molecular distribution, while closure variables in Grad-type and regularised moment systems are higher-order, sign-changing and tail-weighted projections. The latter are especially sensitive to high-peculiar-speed tails, where sign-changing cancellations are almost invisible to density and velocity but dominate fourth-order transport. This distinction is central in kinetic gas theory (Cercignani 1988; Sone 2007; Bird 1994) and in reduced moment models descended from Grad's expansion (Grad 1949; Levermore 1996; Struchtrup 2005; Torrilhon 2016). In the regularised 13- and 26-moment hierarchies, higher moments are not optional diagnostics: the stress equation transports third-order information, and the heat-flux equation transports fourth-order information (Struchtrup & Torrilhon 2003; Torrilhon & Struchtrup 2004; Struchtrup & Torrilhon 2007; Gu & Emerson 2009). In the regularised 26-moment (R26) system, the third-order tensor m_{ijk}, the fourth-order tensor R_{ij}^{cl} and the scalar fourth-order excess Δ are promoted to closure-level variables. A kinetic solver or reduced model that reproduces the visible shock profile but misses these variables has not recovered the closure information needed by the moment hierarchy.

Normal shocks are a particularly stringent test of this issue. They have long served as kinetic benchmarks for the Boltzmann equation, direct simulation Monte Carlo (DSMC), moment closures and model kinetic equations because the shock layer concentrates transport, production and tail populations into a narrow spatial region (Mott-Smith 1951; Alsmeyer 1976; Ivanov & Gimelshein 1998). Recent comparisons of Bhatnagar–Gross–Krook (BGK), Shakhov, ellipsoidal-statistical BGK (ES-BGK) and Fokker–Planck variants confirm that models with similar lower-order transport can differ substantially in higher-order moments in shock waves (Fei *et al.* 2020). This makes the normal shock an appropriate base problem for asking not only whether a solver reproduces low-order fields, but also which closure content its residuals actually identify.

The issue considered in this paper is increasingly relevant because residual-based and physics-informed neural solvers are now used for kinetic equations (Raissi *et al.* 2019; Karniadakis *et al.* 2021; Lou *et al.* 2021; Oh *et al.* 2025). Such solvers are commonly validated using low-order profiles or small residuals in selected moment equations. That practice is potentially misleading in rarefied flows: a residual can be exactly satisfied while observing only a projected part of the nonequilibrium distribution. A companion neural-shock study (Roohi 2026*a*) showed this failure mode empirically: a residual-consistent BGK shock solver could recover the macroscopic profiles and several lower nonequilibrium moments while leaving the tensorial fourth-order closure moment poorly determined. We examine the information content of the kinetic budgets independently of the neural architecture. Specifically, we ask which fourth-order closure content is observable from the heat-flux budget of a normal shock. The answer is structural: the heat-flux budget selects a combined channel, while recovery of the tensorial closure moment itself requires one additional scalar fourth-order channel.

We separate two questions: which fourth-order projection appears in the heat-flux flux, and how collisions produce or relax its scalar-excess component. Deterministic discrete-velocity-method (DVM) references allow us to evaluate budgets without particle-sampling noise. BGK supplies the reference two-budget reconstruction; Shakhov tests the effect of correcting heat-flux relaxation; ES-BGK and ellipsoidal-statistical Fokker–Planck (ES-FP) test scalar-source corrections due to anisotropic targets and diffusion. A separate DSMC calculation measures scalar production directly from accepted binary collisions. The observation channel is kinematic, whereas its complementary scalar-source law depends on the collision operator.

Data-driven moment reduction also addresses the closure problem directly. Han *et al.* (2019) learn generalized moments and their dynamics from kinetic solutions, while Huang *et al.* (2023) enforce global hyperbolicity in a learned radiative-transfer closure. These approaches concern reduced dynamics and structural constraints. Our question is complementary: given the specific heat-flux observation map of a normal shock, which fourth-order components can its residual identify? The rank test below addresses that map, rather than establishing a limitation of every data-driven closure or replacing their realizability and stability constraints.

The gap addressed here is between validation of an observable and identification of its underlying closure variables. Agreement in a projected budget tests that projection; it does not determine components in the observation map's null space. Consequently, comparisons of shock profiles and collision-model source laws, while necessary, do not alone answer the identifiability question posed here. We separate three tasks: identifying the observable channel, supplying independent scalar information, and testing the numerical reliability of the resulting reconstruction. This distinction makes the contribution complementary to established moment hierarchies and kinetic shock benchmarks rather than a replacement for them.

We first derive the heat-flux observation operator and its null space, then construct the complementary scalar-excess budget. The numerical tests examine budget recovery at Mach 2, 3 and 5, direct fourth-order moments at Mach 6, 8 and 12, sparse and noisy scalar measurements, and collision-model source diagnostics. A standalone DSMC calculation accumulates changes in $|c|^4$ over accepted collisions. The contribution is the identification and reconstruction of the tensorial–scalar split within established moment budgets; it introduces neither a new R26 balance law nor a neural solver.

2. Kinetic shocks, R26 moments and DVM references

We use $Pr = \mu c_p/\kappa$, where μ is dynamic viscosity, κ thermal conductivity and $c_p = 5R/2$ the monatomic specific heat at constant pressure. The specific gas constant R is one in computational units; R_{ij}^{cl} instead denotes a fourth-order closure moment. The upstream Mach number is $M_1 = u_1/\sqrt{\gamma RT_1}$, with $\gamma = 5/3$. Subscripts 1 and 2 denote upstream and downstream reservoir states, respectively, and $p = \rho RT$ is the scalar pressure. The third-order tensor m_{ijk} is the symmetric trace-free third central moment of the molecular distribution.

Table 1. Collision model, transport parameters and numerical scope of the separate campaigns. BGK, Shakhov and the separate Mach-2/3 ES-BGK rows are evolved kinetic calculations; ES source-diagnostic rows are evaluations on BGK distributions. The DSMC Prandtl number is a transport consequence of the collision model, rather than an input relaxation-target parameter. Finite-time snapshots and completed prescribed step counts do not certify stationarity.

Campaign	M_1	Pr	Relaxation/transport	Scope
BGK budget references	$2,3,5$	1	Constant $\tau = 1$, $\mu = p\tau$	Full 3V DVM; two-budget recovery
BGK sensitivity / high-Mach extension	5 / $6,8,12$	1	Constant $\tau = 1$	Prescribed-step upwind DVM; exploratory high-Mach moments
Shakhov cross-method DVM	$2,3,4,5$	$2/3$	$\mu = T$, $\tau = 1/\rho$	Reduced-velocity, finite-time profiles
Corrected Shakhov R26	$2,3$	$2/3$	$\mu = T$, $\tau = 1/\rho$	Steady residual solve with independent finite-tolerance qualification
Shakhov published-profile benchmark	8	$2/3$	$\mu = \mu_{\mathrm{ref}} T^{0.81}$	Reduced-velocity upwind DVM; positivity/stationarity limitations
ES-BGK / ES-FP source diagnostics	$2,3,5$	$2/3$	$\alpha_{\mathrm{ES}} = -1/2$, $\beta = -5/4$	Source post-processing on BGK fields
Independent ES-BGK DVM	$2,3$	$2/3$	$\mu = T$, $\tau_{\mathrm{ES}} = 3/(2\rho)$	Evolved finite-time shocks; appendix G
DSMC collision-production ensemble	3	Model-derived	Hard-sphere collisions	Eight-seed original/corrected sampler comparisons
DSMC high-particle profiles	$2,3,4,5$	Model-derived	variable hard sphere (VHS) $\omega = 1$, isotropic scattering	One seed per Mach; finite-time sampling

The calculations comprise several distinct solver campaigns. The Prandtl number is selected by the collision model and its parameters, not by the Mach number. In particular, the BGK budget references at $M_1 = 2,3,5$ and the BGK moment extension at $M_1 = 6,8,12$ have $Pr = 1$. The separate transport-matched Shakhov DVM profiles at $M_1 = 2,3,4,5$ have $Pr = 2/3$, as do the corrected-Shakhov R26 profiles reported at $M_1 = 2,3$. The separate published-profile Mach-8 Shakhov benchmark also uses $Pr = 2/3$, but has a different viscosity law. Table 1 identifies these campaigns before their results are compared. The monatomic value $2/3$ corrects heat-flux relaxation relative to viscous relaxation; it does not make Shakhov, R26 and DSMC collision production identical.

We begin with the steady one-dimensional BGK model (Bhatnagar *et al.* 1954),

$$v_x \partial_x f = \frac{1}{Kn}\{M[f] - f\}, \tag{2.1}$$

Here $Kn = \tau v_0 / L_0$ is the dimensionless relaxation coefficient. We choose $v_0 = \sqrt{RT_1}$ and, for the BGK budget profiles, $L_0 = \lambda_{\mathrm{rel},1} = v_0 \tau_1$, the upstream relaxation length.

With constant $\tau = \tau_1$, this gives $Kn = 1$ and the collision update $\exp(-\Delta t)$ in computational units. Here $Kn = 1$ follows from choosing the upstream relaxation length as the reference length; it does not mean that the mean free path equals the full computational-domain length. The factor $\exp(-\Delta t)$ expresses relaxation over a time step measured in units of τ_1. This choice of units does not itself determine numerical accuracy, which must be assessed through spatial, velocity-space and time-step resolution. The label x/λ_1 on these BGK plots denotes $x/\lambda_{\mathrm{rel},1}$; it does not identify this length with a hard-sphere mean free path. A domain-scaled coordinate, $\texttt{x_scaled} = \texttt{x_mfp}/(2H)$, and coefficient $\texttt{kn_eff} = 1/(2H)$ are also stored by the driver, where H is the half-domain length in relaxation units. The reported BGK budgets use the unscaled relaxation-length coordinate and unit source rate. The Shakhov and DSMC campaigns specify their distinct transport and spatial conventions below.

Here $f(x,v)$ is the distribution and $M[f]$ is the local Maxwellian with the same density ρ, streamwise velocity u_x and temperature T as f. The peculiar velocity is $c = v - u_x e_x$. The heat flux and normal stress are

$$q_x = \frac{1}{2}\int |c|^2 c_x f \,\mathrm{d}v, \qquad \sigma_{xx} = \int \left(c_x^2 - \frac{|c|^2}{3}\right) f \,\mathrm{d}v. \tag{2.2}$$

The coordinate x is normal to the shock, v is the molecular velocity and e_x the streamwise unit vector; $c_x = v_x - u_x$. Distributions are mass-weighted, so $\rho = \int f \,\mathrm{d}v$, $\rho u_x = \int v_x f \,\mathrm{d}v$ and $3\rho T = \int |c|^2 f \,\mathrm{d}v$ in units $R = 1$. Tensor indices range over x, y, z, repeated Cartesian indices are summed, and δ_{ij} is the Kronecker delta. The full deviatoric stress is $\sigma_{ij} = \int (c_i c_j - |c|^2 \delta_{ij}/3) f \,\mathrm{d}v$; σ denotes the matrix of these entries. The fourth-order quantities of interest are

$$R_{xx}^{\mathrm{cl}} = \int |c|^2 \left(c_x^2 - \frac{|c|^2}{3}\right)(f - M_d)\,\mathrm{d}v - 7T\sigma_{xx}, \tag{2.3}$$

The superscript cl identifies the tensorial closure moment. The subtraction $7T\sigma_{xx}$ removes the conventional stress contribution to the contracted fourth moment; its Gaussian-moment derivation is given in appendix F. With M_d the conservative discrete Maxwellian on the same velocity grid, the scalar excess is

$$\Delta_d = \int |c|^4 (f - M_d)\,\mathrm{d}v. \tag{2.4}$$

The continuum-reference excess is $\Delta_c = \int |c|^4 f \,\mathrm{d}v - 15\rho T^2$, whereas Δ_d subtracts the fourth moment of M_d on the actual grid. Their difference is the local equilibrium quadrature offset, which need not be spatially constant. Likewise, the continuum-reference tensorial moment omits the M_d subtraction in (2.3). The heat-budget implementation uses continuum-reference moments; our DVM sensitivity and high-Mach extractions use discrete-reference moments without plateau subtraction. If display offsets $o_R(x)$ and $o_\Delta(x)$ are removed, the corresponding channel offset is $o_R + o_\Delta/3$ and must also be retained in any flux or source reconstruction. The regenerated budget figures use continuum-reference moments with the documented plateau subtraction; their distribution inputs, discrete-source evaluation and plotting scripts are recorded in the deposited reproducibility report.

Table 2. Guide to the moments and their roles in the present analysis.

Quantity	Definition and interpretation
q_x	$\frac{1}{2}\int \lvert c\rvert^2 c_x f\,\mathrm{d}v$: third-order heat transport.
σ_{xx}	$\int (c_x^2 - \lvert c\rvert^2/3) f\,\mathrm{d}v$: deviatoric second-order stress.
R_{xx}^{cl}	Equation (2.3): tensorial fourth-order closure after the lower-order contribution is removed.
Δ_d	Equation (2.4): scalar fourth-order excess relative to the discrete Maxwellian.
S	Equation (3.2): the composite fourth-order channel observed by the one-dimensional heat-flux budget.
Q_Δ	Collision production of the scalar fourth-order moment; its source law depends on the collision model.

The upstream state is normalised by $\rho_1 = T_1 = 1$ and $u_1 = M_1\sqrt{\gamma T_1}$ with $\gamma = 5/3$. The downstream state is set by the monatomic Rankine–Hugoniot relations. The spatial origin is test-specific: aligned ensemble and cross-method comparisons use $(x - x_s)/\lambda_1$, with x_s the density-midpoint location; archived single-run budget profiles retain their stored x/λ_1 coordinate. The overview of the closure moments retains its maximum-density-gradient centring. The captions distinguish these conventions. The Mach-2, Mach-3 and Mach-5 DVM references, generated with deterministic velocity-space quadrature in the spirit of discrete-velocity kinetic solvers (Mieussens 2000; Filbet & Jin 2010; Dimarco & Pareschi 2014), use domains $[-40,40]$, $[-60,60]$ and $[-80,80]$, with $N_x = 1600, 1800, 2200$ spatial points and tensor-product velocity grids $97\times19\times19$, $121\times23\times23$ and $141\times27\times27$ for $M_1 = 2,3,5$, respectively. Their symmetric velocity-box bounds are $v_{\max} = 12,16,22$ in units of $\sqrt{RT_1}$, with composite-trapezoidal quadrature. The archived solver family uses first-order upwind streaming and $\Delta t = C_{\mathrm{CFL}}\Delta x/v_{\max}$, with a default $C_{\mathrm{CFL}} = 0.75$, a conservative discrete-Maxwellian target, exponential relaxation, incoming Rankine–Hugoniot Maxwellian reservoirs, periodic recentering and a 10^{-30} floor. The available reference files do not retain their completed step counts or a residual-based stopping record; we therefore do not assign the later Mach-5 sensitivity campaign's 52000-step count to these references or identify the two Mach-5 inputs as the same run. Full distributions are stored for the budget evaluations. A fixed laboratory-frame velocity box must cover the upstream and downstream populations, but does not adaptively resolve bimodality. Appendix A reports separate Mach-5 extent and refinement tests; resolved outer-box moments do not measure the omitted tails. The separate high-Mach BGK extension uses $M_1 = 6,8,12$, with the full and lightweight configurations listed in table 3; its direct moment comparisons are reported in §5.4. The spatial budget reconstructions in §4–5.3 use the Mach-2, Mach-3 and Mach-5 BGK DVM references because they provide a clean relaxation source for both the heat-flux and scalar-excess budgets. Collision-model robustness is examined through the source and channel diagnostics and the separate evolved Mach-2/3 ES-BGK campaign in appendix G: the Shakhov target changes the heat-flux relaxation but leaves the scalar-excess source neutral, while the ES-BGK and ES-FP targets produce stress-dependent scalar-excess corrections through, respectively, an anisotropic Gaussian covariance and anisotropic Fokker–Planck diffusion.

Table 3. High-Mach constant-relaxation BGK configurations. The stored spatial coordinate spans $[-L,L]$ in the upstream relaxation-length coordinate convention used in the figures; velocities are in units of $\sqrt{RT_1}$, with $R = T_1 = 1$. Each velocity component spans $[-v_{\max}, v_{\max}]$. Counts include the endpoints. "Full" and "lightweight" identify the two stored configurations, rather than a convergence ranking.

Case	L	N_x	$N_{v_x} \times N_{v_y} \times N_{v_z}$	$v_{\max}$	Δx	Δv_x
M6 full	95	2400	$151 \times 29 \times 29$	26	0.07920	0.34667
M8 full	120	2800	$171 \times 31 \times 31$	32	0.08574	0.37647
M12 full	240	4200	$221 \times 35 \times 35$	64	0.11431	0.58182
M8 lightweight	120	2400	$151 \times 29 \times 29$	32	0.10004	0.42667
M12 lightweight	180	3200	$181 \times 31 \times 31$	56	0.11254	0.62222

Here N_x and N_{v_i} count spatial points and velocity nodes, L is the spatial half-length, $v_{\max}$ the velocity-box half-width, and Δx, Δv_i and Δt denote spacings and the time step. The CFL number is $C_{\mathrm{CFL}} = v_{\max}\Delta t/\Delta x$, and N_t denotes the completed number of time steps. The spatial and velocity grids in table 3 are uniform, with $\Delta x = 2L/(N_x - 1)$ and $\Delta v_i = 2v_{\max}/(N_{v_i} - 1)$. Thus the larger point counts do not imply smaller spacings: both Δx and Δv_x increase from the full Mach-6 to Mach-12 case as their domains expand. The transverse spacings $\Delta v_y = \Delta v_z$ are 1.85714, 2.13333 and 3.76471 for these three full configurations. Domain extent and quadrature resolution are distinct numerical choices.

The enlarged domains accommodate the increasing molecular-speed range of the stronger shocks. For the monatomic Rankine–Hugoniot states,

$$\frac{T_2}{T_1} = \frac{(5M_1^2 - 1)(M_1^2 + 3)}{16M_1^2}, \qquad u_1 = M_1\sqrt{5T_1/3}. \tag{2.5}$$

The downstream temperatures are therefore $T_2/T_1 = 12.1198$, 20.8721 and 45.8737 at Mach 6, 8 and 12. A fixed laboratory-frame velocity box must cover the shifted upstream population and the broader downstream thermal population, whose width scales as $\sqrt{T_2}$. Increasing $v_{\max}$ provides room for these populations; increasing the node counts partly compensates for the wider box. Tail coverage is particularly relevant here because fourth-order kernels weight large peculiar speeds by factors of order $|c|^4$. Matching density, momentum and energy on a finite grid does not by itself control these higher moments.

A separate matched-parameter Mach-8 Shakhov benchmark compares density and temperature with Chen *et al.* (2013) in appendix E; its scope and limitations are stated there. Transport over one relaxation time samples distances of order $|v_x|\tau$; the increasing bulk and thermal speeds therefore motivate greater separation between the shock layer and its equilibrium reservoirs. The larger spatial intervals provide this additional buffer for the non-equilibrium profiles and their tails. This is a rationale for the chosen computational domains, not a universal law for shock thickness or evidence that each boundary is sufficiently remote. The high-Mach full/lightweight comparisons vary several parameters together; isolated spatial-domain, velocity-domain, grid

and temporal convergence remain separate requirements. The independent direct simulation Monte Carlo (DSMC) collision-source test is described, with its full sampling protocol and particle-number audit, in §7.

A separate Mach-5 sensitivity campaign varies velocity-domain extent, velocity resolution, spatial resolution, time step and spatial-domain extent (appendix A). Over the aligned shock-core interval, the largest relative differences among the six tested fields are 0.0242%, 0.3993%, 0.0800% and 0.5257% for velocity extent, spatial refinement, half time step and spatial extent, respectively. The velocity-refinement integration is complete at 52000 steps, with a maximum difference of 0.2057%. In particular, spatial refinement also changes the Courant–Friedrichs–Lewy (CFL)-controlled time step. Source-level ES-BGK and ES-FP diagnostics are not independent full kinetic shock solutions.

3. Observation operator and closure-channel identifiability

The inverse question considered here is whether the heat-flux balance contains enough information to recover two distinct fourth-order moments. We take the lower-order fields and the required collision-source contributions as supplied, and ask what this one balance determines about the remaining moments. This separates an information question from the accuracy of a particular numerical reconstruction.

The choice of unknowns follows the moment hierarchy, rather than an assumed closure relation. In the usual R26 organization, a total of 13 equations for density (continuity equation), velocity (momentum equations in 3 directions), temperature (energy equation), stress (5 independent equations), and heat-flux (3 equations) are supplemented by seven components of the symmetric trace-free third-order moment m_{ijk}, five of the contracted trace-free fourth-order moment R^{cl}_{ij}, and the scalar fourth-order excess Δ (Gu & Emerson 2009). The last two describe different aspects of the fourth-order content: directional anisotropy and scalar excess relative to equilibrium.

For the normal-shock branch considered here, the distribution is invariant under transverse reflections and interchange of the two transverse velocity coordinates. Reflection makes every off-diagonal moment integrand odd, so those components vanish; interchange gives $R^{\mathrm{cl}}_{yy} = R^{\mathrm{cl}}_{zz}$. The tensor is trace-free by its defining kernel, hence $R^{\mathrm{cl}}_{xx} + 2R^{\mathrm{cl}}_{yy} = 0$ and $R^{\mathrm{cl}}_{yy} = R^{\mathrm{cl}}_{zz} = -R^{\mathrm{cl}}_{xx}/2$. Appendix F gives the componentwise proof and its discrete-grid conditions.

These reductions use symmetries in *velocity space*, not merely dependence on one spatial coordinate. A distribution $f(x, c_x, c_y, c_z)$ can vary only with x while having different transverse velocity populations. Dependence on x alone imposes neither $R^{\mathrm{cl}}_{yy} = R^{\mathrm{cl}}_{zz}$ nor vanishing off-diagonal components. Under the stated normal-shock symmetries, one scalar function, $R^{\mathrm{cl}}_{xx}(x)$, therefore specifies this tensor, i.e., the full trace-free tensor $R^{\mathrm{cl}}_{ij}(x)$ is determined by its streamwise component:

$$R^{\mathrm{cl}}(x) = R^{\mathrm{cl}}_{xx}(x)\,\mathrm{diag}(1, -1/2, -1/2).$$

The condition $R^{\mathrm{cl}}_{xx} + R^{\mathrm{cl}}_{yy} + R^{\mathrm{cl}}_{zz} = 0$ relates its three diagonal components only: it contains no Δ and supplies no equation for that separate scalar moment. Remind that Δ is the excess of the scalar moment $\int |c|^4 f \, \mathrm{d}v$, not the trace of R^{cl}_{ij}. The isotropic-mixture

example in appendix F explicitly shows that a zero tensorial moment can coexist with a non-zero scalar excess.

The connection to the heat-flux observation follows from the transport term in the heat-flux balance. The heat flux q_x is a third-order moment with kernel $\psi_q = |c|^2 c_x/2$. The flux transporting this moment in the x direction is

$$F_q = \int v_x \psi_q f \,\mathrm{d}v = u_x q_x + \frac{1}{2} B_{xx}[f], \qquad B_{xx}[f] = \int |c|^2 c_x^2 f \,\mathrm{d}v,$$

since $v_x = u_x + c_x$. Thus F_q, rather than q_x itself, contains a fourth-order moment. Its kernel has the exact decomposition

$$|c|^2 c_x^2 = |c|^2 \left(c_x^2 - \frac{|c|^2}{3} \right) + \frac{|c|^4}{3}.$$

The first term selects the tensorial part and the second selects the scalar part; the factor $1/3$ comes from the three-dimensional velocity-space trace. Using (2.3) and (2.4) therefore gives

$$2(F_q - u_x q_x) - B_{xx}[M_d] - 7T\sigma_{xx} = R_{xx}^{\mathrm{cl}} + \frac{\Delta_d}{3}.$$

This identity selects the inverse unknowns: once the equilibrium and lower-order contributions on the left are supplied, the remaining fourth-order information consists of a weighted sum of R_{xx}^{cl} and Δ_d. To ask whether those two contributions can be recovered separately, we collect them as

$$z(x) = \begin{pmatrix} R_{xx}^{\mathrm{cl}}(x) \\ \Delta(x) \end{pmatrix}. \tag{3.1}$$

Here $\Delta = \Delta_d$ for the discrete-reference identity above; other reference conventions must be applied consistently to both moments and the subtracted terms. The vector z denotes the two unknown contributions to this flux, not the full R26 state. Their combination is the scalar observation

$$\mathscr{H} z = R_{xx}^{\mathrm{cl}} + \Delta/3 \equiv S. \tag{3.2}$$

By observation we mean the combination accessible from the heat-flux balance when its source, lower-order terms and an anchor flux are supplied (§4); knowing q_x alone is not a measurement of S. With continuum-reference moments, $B_{xx}[M] = 5\rho T^2$ replaces the discrete equilibrium contribution. Here $B_{xx}[g]$ denotes the fourth-order moment $\int |c|^2 c_x^2 g \,\mathrm{d}v$ of a distribution g. Thus $B_{xx}[M]$ is its equilibrium value for the local Maxwellian $M[f]$ defined after (2.1), giving $B_{xx}[M] = 5\rho T^2$ on full velocity space. Known display offsets must likewise be carried through the reconstruction. These choices change the subtracted terms, not the coefficients of $\mathscr{H}$.

The matrix $\mathscr{H} = [1, 1/3]$ has one non-zero row and two columns: it maps two scalar unknowns to one scalar datum. Its image is one-dimensional, so its rank is exactly one. The two non-zero coefficients do not provide two independent measurements. Indeed,

increasing R_{xx}^{cl} by η can be cancelled by decreasing Δ by 3η:

$$R_{xx}^{\mathrm{cl}} \to R_{xx}^{\mathrm{cl}} + \eta, \qquad \Delta \to \Delta - 3\eta. \tag{3.3}$$

Since $\mathscr{H}(1,-3)^T = 0$, the unobserved direction is

$$\mathscr{N}(\mathscr{H}) = \mathrm{span}\left\{ \begin{pmatrix} 1 \\ -3 \end{pmatrix} \right\}. \tag{3.4}$$

Here $\mathscr{N}$ denotes the null space and $\eta = \eta(x)$ is a scalar perturbation with the same units as R_{xx}^{cl}. Consequently, agreement with S alone cannot establish that either constituent moment has been recovered correctly.
The differential heat-flux balance has the same limitation when its source and lower-order contributions are supplied: it determines $\partial_x S$, and a consistently evaluated boundary flux fixes the integration constant (§4). A perturbation $\eta(x)$ that vanishes at the boundaries still leaves both S and those boundary values unchanged. This is an ambiguity of the selected observation, not a claim that every such perturbation is a realisable distribution or a solution of the full kinetic or R26 equations. Additional moment equations or constitutive information can constrain the unobserved direction. The flux-side projection is kinematic and persists across collision models; evaluating its source, or obtaining an independent channel, can require model-specific information.

A second scalar observation removes the ambiguity if and only if it is not collinear with $\mathscr{H}$. Suppose an auxiliary channel observes

$$\mathscr{P}z = aR_{xx}^{\mathrm{cl}} + b\Delta. \tag{3.5}$$

Here a and b are supplied scalar weights defining $\mathscr{P}$. Then the pair $(\mathscr{H}, \mathscr{P})$ is invertible exactly when

$$\det \begin{pmatrix} 1 & 1/3 \\ a & b \end{pmatrix} = b - a/3 \neq 0. \tag{3.6}$$

The scalar-excess channel corresponds to $(a,b) = (0,1)$ and supplies the one additional scalar field needed for inversion, yielding directly

$$R_{xx}^{\mathrm{cl}} = S - \Delta/3. \tag{3.7}$$

This is a pointwise inversion; reconstructing an entire field from finitely many samples additionally requires spatial interpolation assumptions (§6). The algebra explains why sparse scalar-excess probes can work: the missing information is a one-field complement to the observed heat-flux projection, not a dense direct measurement of the tensorial closure moment. Invertibility alone does not quantify reconstruction accuracy: with a common dimensional scaling, the perturbations satisfy $\delta R_{xx}^{\mathrm{cl}} = \delta S - \delta\Delta/3$ and hence $\|\delta R_{xx}^{\mathrm{cl}}\| \leq \|\delta S\| + \|\delta\Delta\|/3$. Relative errors can still be large where the recovered tensorial moment is small.
The scalar-excess budget in §5 provides a physical route to this additional information; it is not inferred from the heat-flux balance itself.

This also clarifies the connection to R26. In the multidimensional heat-flux hierarchy, the fourth-order contribution is one half of the divergence of

$$A_{ij} = R^{\mathrm{cl}}_{ij} + \frac{1}{3}\Delta\,\delta_{ij}. \tag{3.8}$$

For a planar normal shock, the streamwise equation contains $\partial_x A_{xx}/2$, which gives the projection (3.2). R26 evolves the tensorial and scalar moments through additional equations, whereas a diagnostic using only the heat-flux balance has discarded that extra information. Our rank statement concerns this restricted diagnostic. In higher dimensions even the full tensor A_{ij} is not directly observed: the heat-flux equations contain its divergence, so recovery also depends on boundary or constitutive information.

Figure 1 distinguishes the heat flux from the fourth-order moments entering its transport balance. The heat flux q_x is a third-order moment, whereas $S = R^{\mathrm{cl}}_{xx} + \Delta/3$ contributes to the flux F_q that transports this moment; see (4.2). Consequently, q_x and S need not have similar spatial profiles. In these shocks, q_x remains negative within the shock layer, while the fourth-order profiles change sign. The bottom-row overlays show how the scalar contribution $\Delta/3$ modifies the tensorial profile: identifying the observed channel S with R^{cl}_{xx} would discard this contribution. The profiles illustrate this distinction; verification of the differential budget is presented separately in figure 2.

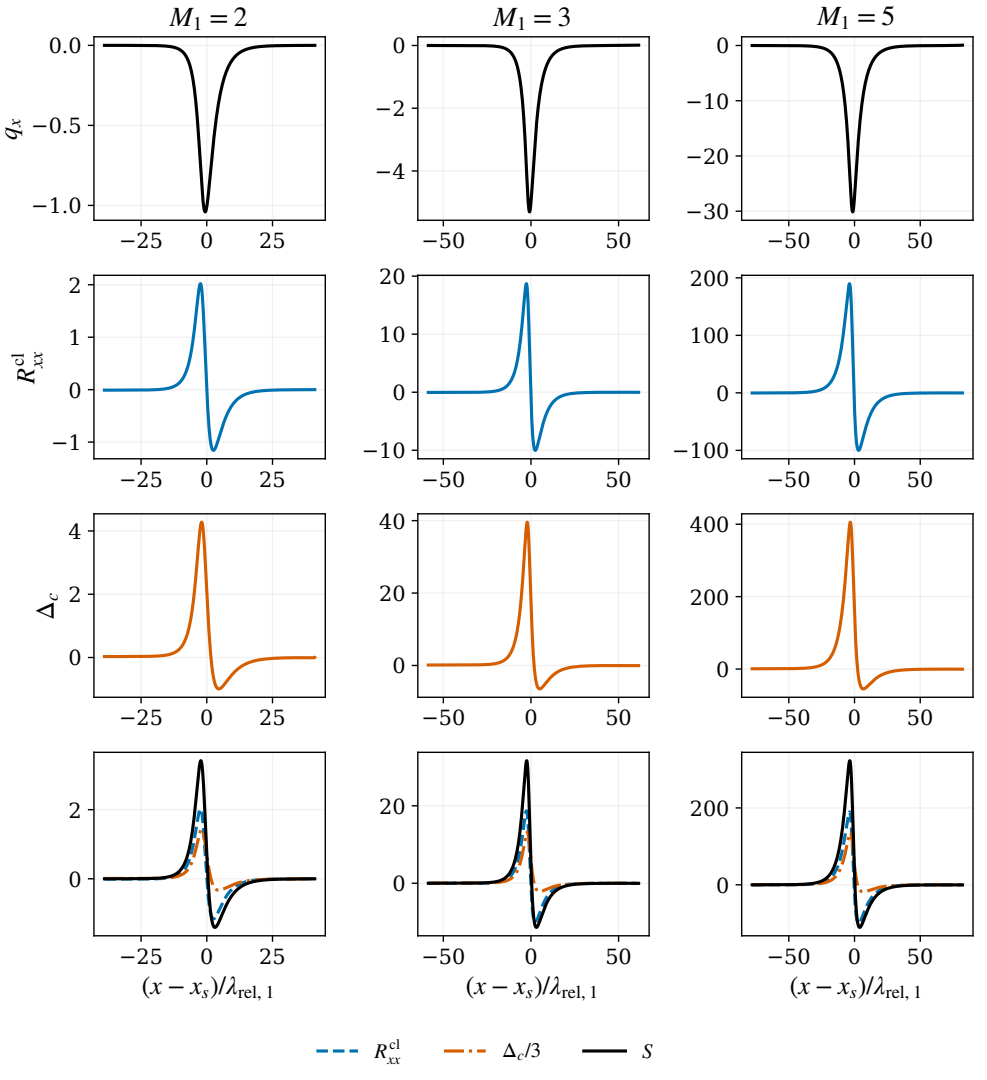


Figure 1. Kinetic normal-shock reference profiles at $M_1 = 2$, 3 and 5. Rows 1–3 show q_x, R_{xx}^{cl} and Δ. The bottom row overlays R_{xx}^{cl} (blue dashed), $\Delta/3$ (orange dash-dotted) and $S = R_{xx}^{\mathrm{cl}} + \Delta/3$ (black solid) on a common vertical scale within each panel. Vertical scales differ between panels. The archived continuum-reference moments are shown without additional plateau subtraction or smoothing. For this figure, the original plotting convention places x_s at the maximum absolute density gradient, rather than the density midpoint.

11 **4. Exact heat-flux balance and numerical recovery of the observed channel**

The identities below state the exact moment balance. The regenerated budget retains the moving-frame contribution and the finite-grid collision source involving $q_x[M_d] - q_x$, and anchors the integrated flux by its directly evaluated value at the first grid point. It uses the same continuum-reference moment definitions and plateau conventions as the scalar-budget reconstruction. The deposited full-distribution checks specify the coordinate, relaxation scale, discrete target and derivative operations; the reported errors refer to this regenerated reconstruction.

For the DVM counterpart of (2.1), with M replaced by M_d, multiplication by $\frac{1}{2}|c|^2 c_x$ and integration over velocity give

$$\partial_x F_q + (\partial_x u_x)\,\mathscr{L}_q = Q_q^{\mathrm{BGK}}, \qquad Q_q^{\mathrm{BGK}} = \frac{q_x[M_d] - q_x[f]}{Kn}. \tag{4.1}$$

Here $F_q = \int v_x \psi_q f \,\mathrm{d}v$, $\psi_q = |c|^2 c_x/2$, and $\mathscr{L}_q = \frac{1}{2}\int v_x(|c|^2 + 2c_x^2) f \,\mathrm{d}v$. The continuum Maxwellian has $q_x[M] = 0$, recovering $Q_q = -q_x/Kn$. On a fixed finite velocity grid, conservation of density, momentum and energy alone does not impose $q_x[M_d] = 0$; its contribution must be evaluated or bounded before using that simplification. The additional term follows from $\partial_x \psi_q = -\frac{1}{2}(\partial_x u_x)(|c|^2 + 2c_x^2)$. It contains moments of order at most three and therefore does not alter the fourth-order observation map. The factor on the right follows from the same $\frac{1}{2}|c|^2 c_x$ kernel used in the definition of q_x. Using the decomposition established in §3, the flux is

$$F_q = u_x q_x + \frac{1}{2}\left[B_{xx}[M_d] + 7T\sigma_{xx} + S_d\right], \qquad S_d = R_{xx}^{\mathrm{cl}} + \Delta_d/3. \tag{4.2}$$

Here the subscript d emphasises that the budget uses the uncorrected discrete-reference moments. The equilibrium and display-offset conventions are those specified in §3; they must be retained consistently when extracting S from the differential balance. Explicitly, supplying Q_q, u_x and $\mathscr{L}_q$ gives

$$F_q(x) = F_q(x_0) + \int_{x_0}^{x} \left[Q_q(\xi) - u_x'(\xi)\,\mathscr{L}_q(\xi)\right] \mathrm{d}\xi. \tag{4.3}$$

Here x_0 is the anchor coordinate, ξ the integration coordinate, and primes on u_x denote spatial derivatives. The flux $F_q(x_0)$ is evaluated from the supplied reference distribution at an anchor point, or from a prescribed full equilibrium distribution when that condition applies. Specifying only incoming reservoir populations does not prescribe the full boundary flux. Equation (4.2) then gives S_d. One must not impose $S_d = 0$ merely because the anchor is called an equilibrium plateau: on a finite grid, $f = M_d$ gives $\Delta_d = 0$ but $R_{xx}^{\mathrm{cl}} = -7T\sigma_{xx}[M_d]$, which can be non-zero.

Figure 2 compares two ways of evaluating the same fourth-order channel, $S = R_{xx}^{\mathrm{cl}} + \Delta/3$, for the Mach-3 BGK reference. The black solid curve, labelled "full distribution", is obtained by directly evaluating the velocity moments defining R_{xx}^{cl} and Δ from the saved DVM distribution $f(x,v)$. It is therefore a moment profile, not a plot of f itself. The orange dashed curve, labelled "integrated heat budget", is obtained through the following sequence. Equation (4.1) supplies the differential balance for F_q.

Equation (4.3) integrates this balance from the first grid point x_0, using the directly evaluated anchor $F_q(x_0)$ and retaining both the finite-grid source $Q_q = q_x[M_d] - q_x[f]$ and the moving-frame term $(\partial_x u_x)\mathscr{L}_q$ in the present computational units. Equation (4.2) then extracts the fourth-order channel from the reconstructed flux. For the continuum-reference moment convention used in this plot, the resulting expression is

$$S_{\mathrm{bud}} = 2\left(F_{q,\mathrm{bud}} - u_x q_x\right) - 5\rho T^2 - 7T\sigma_{xx},$$

with the finite-grid reference offsets treated consistently before plotting. Both curves use the same display convention: the mean over the first and last 8% of the spatial samples is subtracted.

The lower panel evaluates the differential balance using the directly computed flux, rather than its integrated reconstruction. Its normalised residual is

$$\varepsilon_q(x) = \frac{\partial_x F_q - Q_q + (\partial_x u_x)\mathscr{L}_q}{\max_x \left|Q_q - (\partial_x u_x)\mathscr{L}_q\right|}.$$

Thus, the upper panel checks integrated reconstruction of S, whereas the lower panel checks differential consistency of the supplied finite-time profile. The reconstruction uses source and auxiliary moments evaluated from the same DVM distribution; it is not an independent shock solution from low-order fields alone. Agreement does not separately identify R_{xx}^{cl} and Δ, nor does it by itself certify stationarity. Separating these two fourth-order contributions requires the additional scalar budget introduced in §5.

The purpose of this comparison is to check the numerical reconstruction of the observable channel before attempting to recover its individual components. If the directly evaluated F_q were merely substituted into (4.2), agreement would follow from the moment identity. Here, however, the interior flux is reconstructed by integrating (4.3); the direct flux is used only to supply the anchor at x_0. The comparison therefore tests the combined treatment of the collision source, moving-frame contribution, integration and reference offsets. Both evaluations nevertheless use information from the same DVM distribution, so this is a consistency check rather than independent validation of the kinetic model. Its role in the closure-recovery argument is to establish that S can be reconstructed consistently, while agreement in S alone cannot distinguish R_{xx}^{cl} from Δ. That distinction requires the additional scalar budget.

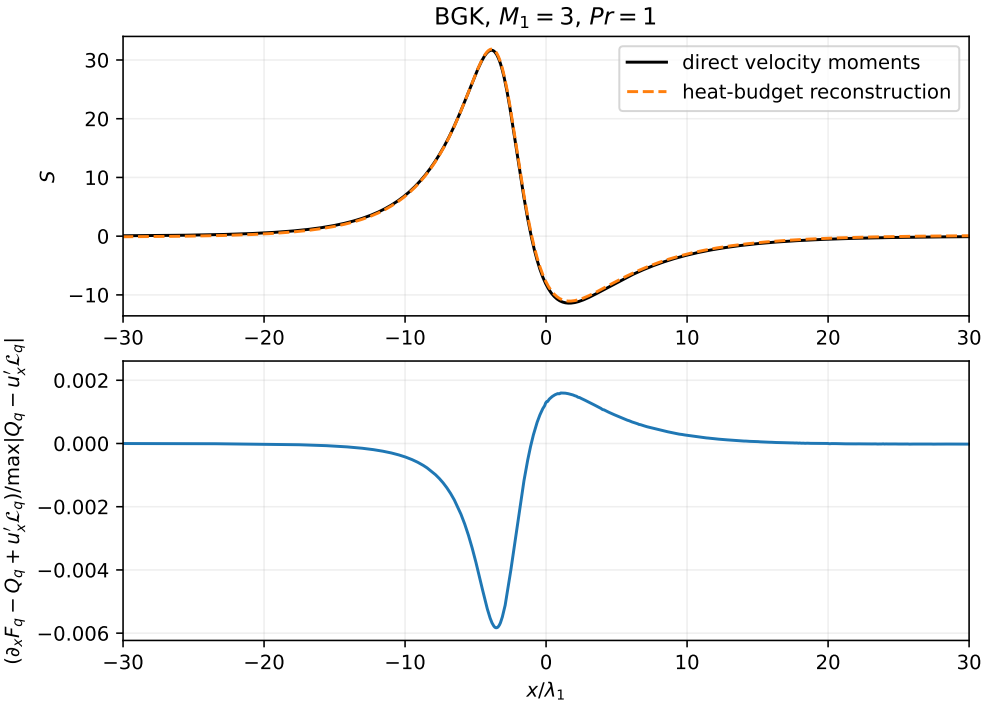


Figure 2. Heat-budget consistency check for the Mach-3 BGK reference ($Pr = 1$). Upper panel: the fourth-order channel $S = R^{\mathrm{cl}}_{xx} + \Delta/3$ evaluated directly from the full DVM distribution (black solid) and reconstructed by integrating the heat-flux balance and extracting S through (4.2) (orange dashed). Both profiles use the same plateau subtraction. Lower panel: the normalised differential residual ε_q, defined in the text, evaluated with the directly computed F_q. Here $Q_q = q_x[M_d] - q_x[f]$ and $\mathscr{L}_q = \int v_x(\frac{1}{2}|c|^2 + c_x^2) f \,\mathrm{d}v$. The symbol A in the plotting data denotes this same $\mathscr{L}_q$; its multiplier is the velocity gradient $\partial_x u_x$, not the velocity u_x. The comparison tests reconstruction and differential consistency, not independent closure recovery or certified stationarity.

5. Scalar-excess budget and two-channel recovery

The missing channel is supplied by a scalar fourth-order budget. Multiplication of (2.1) by $|c|^4$ gives

$$\int |c|^4 v_x \partial_x f \,\mathrm{d}v = Q_\Delta . \tag{5.1}$$

Because the peculiar velocity depends on the local velocity $u_x(x)$, the flux derivative contains a moving-frame contribution. Define

$$J_\Delta = \int v_x |c|^4 f \,\mathrm{d}v, \qquad K_\Delta = \int v_x |c|^2 c_x f \,\mathrm{d}v. \tag{5.2}$$

These fluxes specify what additional information is needed. In fact $K_\Delta = 2F_q$, whereas

$$J_\Delta = u_x \int |c|^4 f \,\mathrm{d}v + \int c_x |c|^4 f \,\mathrm{d}v$$

contains a fifth-order moment not determined by the lower-order fields and S. Here these fluxes are evaluated from the full DVM distribution: the two-budget procedure tests moment recovery and consistency, rather than defining a closed solver from low-

order data alone. Since $\partial_x|c|^4 = -4|c|^2 c_x \partial_x u_x$, one obtains

$$\int |c|^4 v_x \partial_x f \, \mathrm{d}v = \partial_x J_\Delta + 4(\partial_x u_x) K_\Delta . \tag{5.3}$$

For BGK, the collision source is

$$Q_\Delta^{\mathrm{BGK}} = \frac{1}{Kn} \int |c|^4 (M_d - f) \, \mathrm{d}v = -\frac{\Delta_d}{Kn}, \tag{5.4}$$

where in the DVM implementation $M \to M_d$. The identity is exact for the uncorrected discrete excess Δ_d in (2.4); use of a plateau-corrected Δ requires the same correction in the source. Thus the raw budget reconstructs Δ_d. A displayed field $\Delta = \Delta_d - o_\Delta$ instead satisfies $Q_\Delta = -(\Delta + o_\Delta)/Kn$ and requires subtracting o_Δ from the reconstructed excess. Hence

$$\Delta_{d,\mathrm{bud}} = -Kn \left[\partial_x J_\Delta + 4(\partial_x u_x) K_\Delta \right]. \tag{5.5}$$

The scalar-budget derivative uses a centred seven-point boxcar on the moment inputs, with the same convention for the Mach-2, Mach-3 and Mach-5 reconstructions.

The form of the scalar-excess source depends on the collision operator, but the flux-side observation operator does not. For a generic kinetic equation $v_x \partial_x f = \mathscr{C}[f]$, the heat-flux source is

$$Q_q = \int \frac{1}{2} |c|^2 c_x \mathscr{C}[f] \, \mathrm{d}v, \qquad Q_\Delta = \int |c|^4 \mathscr{C}[f] \, \mathrm{d}v. \tag{5.6}$$

The source changes with the collision model, while the observed fourth-order channel in the heat-flux flux remains the same kinematic projection. Recovering Δ from a budget requires a source law invertible for Δ once its other inputs are supplied; a general Boltzmann production measurement alone is not such an algebraic inversion. The analytic Shakhov, ES-BGK and ES-FP source laws below use full velocity-space integrals and the continuum-reference excess $\Delta = \Delta_c$, without plateau subtraction. Discrete-reference diagnostics require the corresponding equilibrium-offset corrections; finite-box Fokker–Planck evaluations also require boundary and discretisation checks. BGK gives (5.4). The Shakhov target changes the heat-flux moment so that $Q_q = -Pr\, q_x/Kn$, but its correction is odd in c and integrates to zero against the even kernel $|c|^4$; therefore $Q_\Delta = -\Delta/Kn$ is unchanged. This parity proof applies on full velocity space (or a domain symmetric in c). A fixed laboratory-frame grid is generally not symmetric about the local u_x; discrete neutrality must be evaluated for the actual target and quadrature rather than inferred from this continuum argument. Thus a correct-Prandtl Shakhov correction alters the heat-flux relaxation without changing the scalar-excess source.

Unlike Shakhov, ES-BGK uses an anisotropic Gaussian target (Holway 1966; Fei *et al.* 2020),

$$G_\Theta(c) = \frac{\rho}{(2\pi)^{3/2} (\det \Theta)^{1/2}} \exp\left[-\frac{1}{2} c_i (\Theta^{-1})_{ij} c_j \right], \tag{5.7}$$

with covariance tensor Θ. The operators det and tr denote determinant and trace. The dimensionless coefficient α_{ES} controls target anisotropy. In the conventional

monatomic ES-BGK construction, $\Theta = TI + \Lambda_{\mathrm{ES}}$, where Λ_{ES} is a trace-free tensor proportional to the stress; in the notation used here one may write $\Lambda_{\mathrm{ES}} = \alpha_{\mathrm{ES}}\sigma/\rho$ with the stress convention defined above. Here T is the local scalar temperature in the adopted units ($R = 1$), and I is the 3×3 identity tensor; thus TI represents the isotropic part of the velocity covariance. The heat-flux observation channel does not change, because it is determined by the left-hand-side flux. The scalar-excess source does change, because the target distribution no longer has the Maxwellian fourth scalar moment.

For any zero-mean Gaussian with covariance Θ,

$$\int |c|^4 G_\Theta \, \mathrm{d}v = \rho \left[(\mathrm{tr}\Theta)^2 + 2\,\mathrm{tr}(\Theta^2) \right]. \tag{5.8}$$

This identity follows from Wick's theorem: the brackets here denote expectation under the normalised Gaussian G_Θ/ρ; $\langle c_i c_i c_j c_j \rangle = (\mathrm{tr}\Theta)^2 + 2\,\mathrm{tr}(\Theta^2)$. For a Maxwellian, $\Theta = TI$, so $(\mathrm{tr}\Theta)^2 = 9T^2$ and $2\,\mathrm{tr}(\Theta^2) = 6T^2$, giving $15T^2$. For ES-BGK, however, the trace-free part Λ_{ES} gives

$$\Delta_G^{\mathrm{ES}} = \rho \left[(\mathrm{tr}\Theta)^2 + 2\,\mathrm{tr}(\Theta^2) - 15T^2 \right] = 2\rho\ \mathrm{tr}(\Lambda_{\mathrm{ES}}^2) = \frac{2\alpha_{\mathrm{ES}}^2}{\rho}\,\mathrm{tr}(\sigma^2). \tag{5.9}$$

Thus the ES-BGK scalar-excess source is not simply $-\Delta/Kn$, but

$$Q_\Delta^{\mathrm{ES-BGK}} = \frac{1}{Kn}\left(\Delta_G^{\mathrm{ES}} - \Delta\right). \tag{5.10}$$

In a one-dimensional normal shock, $\sigma_{yy} = \sigma_{zz} = -\sigma_{xx}/2$, so $\mathrm{tr}(\sigma^2) = 3\sigma_{xx}^2/2$ and

$$\Delta_G^{\mathrm{ES}} = \frac{3\alpha_{\mathrm{ES}}^2}{\rho}\sigma_{xx}^2. \tag{5.11}$$

This term is nonnegative: the ES-BGK target carries a non-zero scalar fourth-order excess whenever the covariance is anisotropic. Shakhov changes the odd heat-flux channel and leaves the even $|c|^4$ channel neutral; ES-BGK changes the even scalar-excess target through a stress-squared covariance effect.

To estimate the size of this correction in the present shocks, we post-process the Mach 2, 3 and 5 DVM profiles using (5.11) with $Pr = 2/3$ and $\alpha_{\mathrm{ES}} = 1 - 1/Pr = -1/2$. Table 4 reports the relative magnitude $\|\Delta_G^{\mathrm{ES}}\|/\|\Delta\|$ of the stress-dependent target excess. Here "correction" denotes the additional contribution $\Delta_G^{\mathrm{ES}}/Kn$ to the BGK scalar-excess source $-\Delta/Kn$. The ratio $\|\Delta_G^{\mathrm{ES}}\|/\|\Delta\|$ in the active shock layer is approximately 4.3–4.5% for all three Mach numbers, and the ES-BGK scalar-excess source remains almost collinear with the BGK/Shakhov source, with correlations exceeding 0.999. The correction is therefore modest on these evaluated BGK profiles, but it is systematically non-zero and grows in absolute magnitude with Mach number: the peak value of Δ_G^{ES} increases from 0.196 at $M_1 = 2$ to 20.25 at $M_1 = 5$. These percentages describe source evaluations on the stored BGK profiles, not independently evolved ES-BGK fields. We additionally implement and evolve the ES-BGK kinetic model itself at Mach 2 and 3, with $Pr = 2/3$ and transport matched to the Shakhov campaign. Appendix G reports its solver, finite-time profiles and two-budget reconstructions. The source-only comparison in table 4 and figure 3 is retained

as a controlled comparison on common input fields; ES-FP remains a post-processing diagnostic.

Fokker–Planck models provide a complementary example because their sources are generated by drift and diffusion rather than by relaxation to a target distribution. For a velocity-space Fokker–Planck operator

$$\mathscr{C}_{\mathrm{FP}}[f] = -\partial_{c_i}(A_i f) + \frac{1}{2}\partial_{c_i}\partial_{c_j}(D_{ij} f), \tag{5.12}$$

two integrations by parts on full velocity space, with sufficiently decaying distributions and vanishing boundary terms, transfer the derivatives to $\phi = |c|^4$. Since $\partial_{c_i}\phi = 4|c|^2 c_i$ and $\partial_{c_i}\partial_{c_j}\phi = 8c_i c_j + 4|c|^2\delta_{ij}$, the scalar production is

$$Q_\Delta^{\mathrm{FP}} = \int \left[4A_i|c|^2 c_i + 4D_{ij}c_i c_j + 2D_{ii}|c|^2\right] f \,\mathrm{d}v. \tag{5.13}$$

This form is useful because it makes the difference from BGK-type relaxation explicit. In an ES-FP model with standard linear drift $A_i = -\nu c_i$ and anisotropic diffusion $D_{ij} = 2\nu(T\delta_{ij} + \beta\sigma_{ij}/\rho)$, substitution gives

$$Q_\Delta^{\mathrm{ES-FP}} = \nu\left[-4\Delta + \frac{8\beta}{\rho}\,\mathrm{tr}(\sigma^2)\right] = 4\nu\left[-\Delta + \frac{2\beta}{\rho}\,\mathrm{tr}(\sigma^2)\right]. \tag{5.14}$$

Here A_i is the velocity-space drift, D_{ij} the diffusion tensor, $\nu > 0$ the collision-rate scale and β the dimensionless diffusion-anisotropy coefficient. The fourth-order test function is $\phi(c) = |c|^4$.

For the correct-Prandtl choice $\beta = -5/4$, the ES-FP correction is therefore proportional to the same stress-squared invariant as the ES-BGK correction, but with a different coefficient and opposite sign. For clarity, substitution of $\int c_i c_j f\,\mathrm{d}v = \rho T\delta_{ij} + \sigma_{ij}$ and $\mathrm{tr}\,\sigma = 0$ gives

$$\frac{Q_\Delta}{\nu} = -4(\Delta + 15\rho T^2) + 24\rho T^2 + 36\rho T^2 + \frac{8\beta}{\rho}\,\mathrm{tr}(\sigma^2).$$

The equilibrium terms cancel, yielding (5.14) before any normal-shock symmetry reduction. On a finite velocity box the discarded boundary terms require a separate check. This formal source identity presupposes an admissible diffusion tensor. With $\mathsf{C} = \rho^{-1}\int ccf\,\mathrm{d}v = TI + \sigma/\rho$ the actual distribution covariance, its normalised form is $D/(2\nu) = (1-\beta)TI + \beta\mathsf{C}$. At $\beta = -5/4$, positive definiteness requires every eigenvalue of C to be smaller than $9T/5$. A fixed correct-Prandtl coefficient does not guarantee this condition for arbitrary nonequilibrium states. Entropy inequalities for Gaussian ES-BGK and admissible ES-FP models are established in the literature (Andries *et al.* 2000; Mathiaud & Mieussens 2016). In particular, the latter allows a state-dependent anisotropy parameter to maintain admissible diffusion; a fixed $\beta = -5/4$ cannot be extended to arbitrary far-from-equilibrium states. Those results are distinct from our sampled finite-grid entropy diagnostic. This is the scalar-excess analogue of the high-order-moment production differences emphasised in shock-wave benchmark studies of BGK, Shakhov, ES-BGK, ES-FP and Cubic-FP models (Fei *et al.* 2020). It also explains why equal Prandtl number does not imply equal fourth-order closure production: the heat-flux relaxation rate is a lower-order transport constraint, whereas Q_Δ samples an even fourth-order kernel.

Table 4. Source-level ES-BGK and ES-FP scalar-excess corrections evaluated on the DVM shock profiles. The ratios are active-zone norms relative to $\|\Delta\|$. The opposite signs reflect different mechanisms: anisotropy raises the fourth scalar moment of the ES-BGK Gaussian target, whereas the ES-FP correction contracts the stress with anisotropic diffusion. At $Pr = 2/3$, $\beta/\alpha_{\mathrm{ES}}^2 = -5$; this is a source coefficient ratio, not an entropy-production sign. The ES-FP source is the scaled source $Q_\Delta^{\mathrm{ES-FP}}/(4\nu)$ in (5.14); the correlation compares the full ES-FP scaled source with the BGK/Shakhov baseline $-\Delta$.

Mach	ES-BGK correction	ES-FP correction	$\mathrm{corr}(Q_\Delta^{\mathrm{ES-FP}}, -\Delta)$
M2	0.0431	0.215	0.987
M3	0.0447	0.224	0.984
M5	0.0439	0.220	0.984

The constant source-correction ratio follows directly from this algebra, not from a separate numerical fit. After the ES-FP source is scaled by 4ν, its stress-quadratic correction is

$$Q_{\Delta,\mathrm{corr}}^{\mathrm{ES-FP}}/(4\nu) = \frac{2\beta}{\rho}\,\mathrm{tr}(\sigma^2), \tag{5.15}$$

whereas the ES-BGK correction from (5.9) is

$$\Delta_G^{\mathrm{ES}} = \frac{2\alpha_{\mathrm{ES}}^2}{\rho}\,\mathrm{tr}(\sigma^2). \tag{5.16}$$

Equivalently, $Q_{\Delta,\mathrm{corr}}^{\mathrm{ES-FP}}/(4\nu) = (\beta/\alpha_{\mathrm{ES}}^2)\Delta_G^{\mathrm{ES}}$ for $\alpha_{\mathrm{ES}} \neq 0$. Wherever $\Delta_G^{\mathrm{ES}} \neq 0$, their pointwise ratio is

$$\frac{Q_{\Delta,\mathrm{corr}}^{\mathrm{ES-FP}}/(4\nu)}{\Delta_G^{\mathrm{ES}}} = \frac{\beta}{\alpha_{\mathrm{ES}}^2}. \tag{5.17}$$

For $Pr = 2/3$, $\alpha_{\mathrm{ES}} = 1 - 1/Pr = -1/2$ and $\beta = -5/4$, so (5.17) gives exactly -5. The ratio -5 is therefore an analytical consequence of the ES-BGK Gaussian covariance and ES-FP anisotropic diffusion coefficients; the DVM post-processing only measures the size of the common stress-squared invariant relative to the actual shock-layer Δ.

Table 4 and figure 3 compare the ES-BGK and ES-FP source-level diagnostics on the same DVM shock profiles. The ES-FP correction contributes 21.5%, 22.4% and 22.0% of $\|\Delta\|$ for Mach 2, 3 and 5, respectively, about five times the ES-BGK correction in magnitude and with the opposite sign for $\beta = -5/4$. Nevertheless, the full scaled ES-FP source remains strongly aligned with the BGK/Shakhov scalar-excess source, with correlations 0.987, 0.984 and 0.984. Thus ES-FP does not change the kinematic observation channel; it changes the source that drives the missing scalar complement. This distinction is central to the paper: S is the channel observed by the heat-flux flux, while Q_Δ is the collision-model-specific mechanism by which the scalar-excess information is produced.

This result is consistent with benchmark studies showing that correct-Prandtl kinetic models can agree in low-order transport while differing in higher-order moments and their production terms in shock waves (Fei *et al.* 2020). In the present language, ES-BGK changes the scalar-excess source through a stress-quadratic target term, ES-FP changes it more strongly through anisotropic diffusion, and Shakhov does not change it at all. These differences are invisible if one looks only at density, velocity, stress or heat-flux relaxation

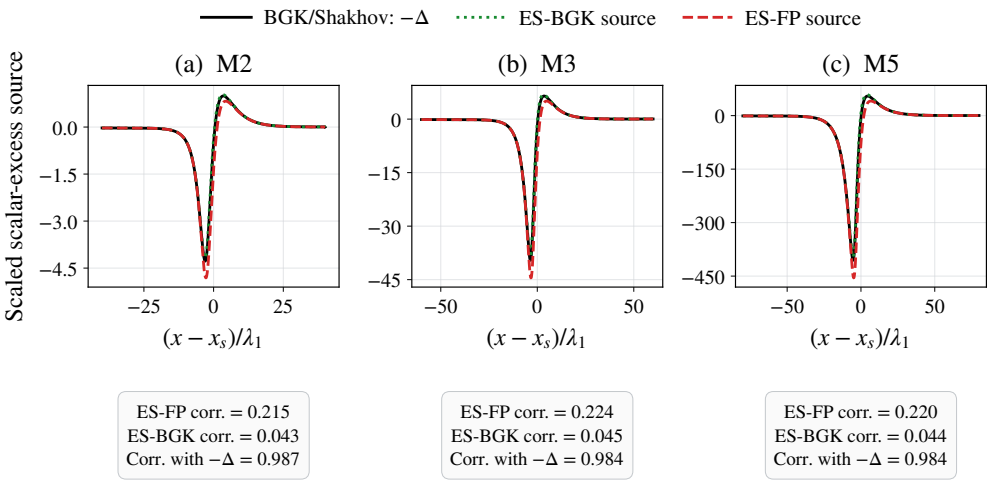


Figure 3. Scaled scalar-excess source for BGK/Shakhov, ES-BGK and ES-FP, evaluated on the Mach 2, 3 and 5 DVM shock profiles. The black curve is the BGK/Shakhov baseline $-\Delta$. ES-BGK gives a small stress-quadratic correction from the anisotropic Gaussian target, whereas ES-FP gives a larger anisotropic-diffusion correction of opposite sign. The nearly coincident shapes show that the source remains strongly aligned with the scalar-excess channel, while the separation between green and red curves shows that correct-Prandtl kinetic models can assign different fourth-order production laws to the same observed heat-flux channel.

rates. They become visible only after the scalar-excess channel has been identified as the missing complement to the heat-flux observation channel.

5.1. *Shakhov source-neutrality and channel-consistency check*

The Shakhov model is the most direct correct-Prandtl test of the BGK result because it changes the heat-flux relaxation without replacing the Maxwellian covariance by an anisotropic one. We therefore evaluate a conservative discrete Shakhov target (Shakhov 1968) on the Mach-3 DVM shock. The Shakhov target, denoted M_S, has the form of a Maxwellian multiplied by an odd heat-flux correction. Consequently the odd correction contributes to the heat-flux projection but vanishes against the even scalar kernel $|c|^4$; the target retains the Maxwellian scalar fourth moment. Numerically, the target gives $q_x[M_S]/q_x[f] = 0.333323 \simeq 1 - Pr$ and the direct heat-flux collision source gives $Q_q/q_x[f] = -0.666677 \simeq -Pr$, as required for $Pr = 2/3$. At the same time, the evaluated scalar-excess source remains $Q_\Delta = -\Delta/Kn$ to numerical precision for this target and grid; this is a numerical check of the discrete construction.

This check is deliberately not another spatial budget reconstruction. It evaluates the collision projections directly from the discrete distribution and therefore isolates the channel identity from the finite-difference derivative error present in figures 5 and 7. Figure 4 shows that the heat-flux flux reconstructs S with a 0.064% active-zone error, the scalar-excess channel reconstructs Δ to numerical precision, and the two-channel recovery reduces the R_{xx}^{cl} error from 68.4% to 0.10%. This source-neutrality verification shows that the heat-flux observation channel and scalar-excess

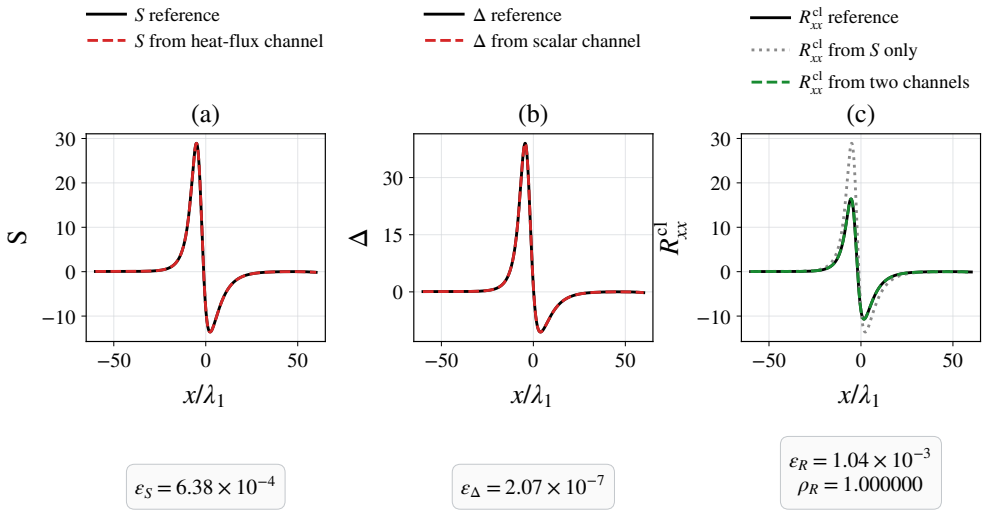


Figure 4. Conservative discrete Shakhov channel check for the Mach-3 shock with $Pr = 2/3$. (a) The heat-flux flux channel reconstructs S. (b) The scalar-excess source is neutral under the Shakhov correction and recovers Δ. (c) The heat-flux channel alone gives a large R_{xx}^{cl} error, whereas the two-channel reconstruction recovers the tensorial closure moment. The panel errors are relative norms expressed as fractions. The collision sources are evaluated directly from the discrete distribution, so this checks channel consistency rather than replacing the BGK spatial budget sweep in figures 5 and 7.

complement survive correction of the Prandtl-number defect of BGK; it does not introduce a new recovery method.

The check uses $Pr = 2/3$ and a unit computational source scale. The distribution minimum is near 10^{-30}, while the signed target minimum is approximately -9.96×10^{-5}; the collision projection therefore does not certify a nonnegative Shakhov target or a stationary shock. Each displayed reference and reconstructed channel has the mean of the first and last 8% of points removed, with active mask $|S| > 0.02 \max |S|$. The agreement in the scalar channel is plateau-corrected: $\Delta_{\text{source}} + KnQ_\Delta$ is a constant offset, approximately 0.06816. The flux-side subtraction uses $5\rho T^2$, whereas the directly evaluated closure moments use $f - M_d$, so the small flux-channel discrepancy also contains the discrete Maxwellian reference defect. The numerical curves have been regenerated from the archived full distribution; the input, code, profiles and provenance are deposited with the article.

A separate short diagnostic compares clipped and signed Shakhov updates for 20 sweeps from the same initial state. Relative differences in R, Δ and S are approximately 1.335%, 1.547% and 1.465%. This finite-sweep comparison does not establish a steady solution for either branch, but it prevents treating positivity clipping as numerically neutral. The analytic odd-parity source-neutrality statement concerns the unmodified collision correction; it must be distinguished from conservation and source balances after clipping.

Table 5. Observation channel and fourth-order source laws for the models considered here. The heat-flux observation channel is kinematic, whereas the source laws are collision-model dependent. The analytic relaxation formulas use continuum-reference moments on full velocity space; discrete equilibrium and quadrature corrections are as described in the text. Here $\psi_q = \frac{1}{2}|c|^2 c_x$, G_Θ is the ES-BGK anisotropic Gaussian, A_i and D_{ij} are Fokker–Planck drift and diffusion coefficients, and $Q(f,f)$ is the Boltzmann collision operator.

Model	Heat-flux source Q_q	Scalar-excess source Q_Δ	Interpretation
BGK	$-q_x/Kn$	$-\Delta/Kn$	Baseline relaxation of heat flux and scalar excess.
Shakhov	$-Pr\,q_x/Kn$	$-\Delta/Kn$	Corrects heat-flux relaxation; odd Shakhov correction is neutral in the even $\lvert c\rvert^4$ channel.
ES-BGK	target-dependent	$(\Delta_G^{\mathrm{ES}} - \Delta)/Kn$	Anisotropic covariance adds a stress-quadratic scalar-excess target.
ES-FP	drift/diffusion dependent	$\nu[-4\Delta + 8\beta\,\mathrm{tr}(\sigma^2)/\rho]$	Anisotropic diffusion changes the scalar-excess production even when the Prandtl number is corrected.
Boltzmann/DSMC	$\int \psi_q Q(f,f)\,\mathrm{d}v$	$\int \lvert c\rvert^4 Q(f,f)\,\mathrm{d}v$	True collisional production; sampled in DSMC from pre/post-collision changes of $\lvert c\rvert^4$.

5.2. *Boltzmann limit and direct DSMC production*

For the Boltzmann collision operator, the analogue is the genuine collision-production integral $\int |c|^4 Q(f,f)\,\mathrm{d}v$. In DSMC this production can be accumulated directly by summing the change of $|c|^4$ over accepted binary collisions. The hierarchy of sources is therefore: BGK and Shakhov give a scalar relaxation of Δ; ES-BGK adds a stress-squared target excess; ES-FP modifies the same channel through anisotropic diffusion; and the Boltzmann operator defines the fourth-order production, which DSMC estimates from molecular collisions. In all cases, however, the flux-side heat-flux observation operator remains the same composite fourth-order channel. The DSMC test in §7 closes this loop by measuring the scalar-excess production directly in a stochastic binary-collision shock.

Table 5 separates the common kinematic channel S from the model-dependent scalar production Q_Δ. This distinction allows the same observation map to be tested with different collision-source laws.

5.3. *Numerical two-budget recovery across Mach number*

The closure recovery now follows directly. The heat-flux budget gives S, the scalar-excess budget gives Δ_{bud}, and the tensorial closure is reconstructed as

$$R^{\mathrm{cl}}_{xx,\mathrm{bud}} = S_{\mathrm{bud}} - \Delta_{\mathrm{bud}}/3. \tag{5.18}$$

The numerical budgets use the actual discrete velocity weights and retain the Maxwellian target moments in the collision contractions. In particular, $Q_q = (q_x[M_d] - q_x[f])/\tau$ and the scalar reconstruction is

$$\Delta_{\mathrm{bud}} = \sum_k w_k |c_k|^4 M_{d,k} - 15\rho T^2 - \tau\left(\frac{\mathrm{d}J_\Delta}{\mathrm{d}x} + 4\frac{\mathrm{d}u_x}{\mathrm{d}x}K_\Delta\right). \tag{5.19}$$

Here k indexes the discrete three-dimensional velocity nodes v_k, w_k denotes their velocity-quadrature weights, and $c_k = v_k - u_x e_x$ is the peculiar velocity at node k, with e_x the unit vector in the shock-normal direction. The quantity $M_{d,k}$ is the conservative discrete Maxwellian target evaluated at that node, matching the density, momentum and energy of the local discrete distribution. The fields ρ, u_x and T are the local density, streamwise mean velocity and temperature, respectively; temperature is expressed in the computational convention $R = 1$. The parameter τ is the BGK relaxation time, equal to one in the computational units used for these budget reconstructions. The two fluxes are evaluated with the same quadrature:

$$J_\Delta = \sum_k w_k v_{x,k} |c_k|^4 f_k, \qquad K_\Delta = \sum_k w_k v_{x,k} |c_k|^2 c_{x,k} f_k,$$

where $f_k = f(x, v_k)$.

Thus Δ_{bud} reconstructs the continuum-reference scalar excess $\Delta_c = \sum_k w_k |c_k|^4 f_k - 15\rho T^2$, before the subsequent plateau subtraction.

Suffixes d and c refer to discrete and continuum equilibrium, respectively; bud, full and sparse identify budget reconstruction, complete scalar-field input and interpolated probe input. Macroscopic fields are recomputed consistently from the stored distribution, and the conservative discrete target is evaluated in double precision with four matching iterations. The heat-flux integral is anchored by its directly evaluated value at the first grid point. With the stored `x_mfp` coordinate, computational $\tau = 1$ is used. Derivatives employ one $(1,4,6,4,1)/16$ pass on u_x for the heat budget and a seven-point boxcar on J_Δ, u_x, K_Δ for the scalar budget. The raw discrete-source inversion reconstructs Δ_d. Adding the local offset $\sum_k w_k |c_k|^4 M_{d,k} - 15\rho T^2$ converts it to Δ_c before the stated plateau subtraction. The heat-flux channel subtracts the continuum term $5\rho T^2$; it retains the directly evaluated $q_x[M_d]$ in the source and the measured first-point flux anchor. The tensorial reference likewise omits the discrete-Maxwellian fourth-moment subtraction. These choices put both reconstructed channels and their reference on the same continuum-reference convention. These budget checks do not establish stationarity or independent grid convergence.

Error measures and comparison sets. For a reconstruction g_{rec} and reference g_{ref} on a set $\mathscr{A}$, we report

$$\varepsilon_g = 100\frac{\|g_{\mathrm{rec}} - g_{\mathrm{ref}}\|_{2,\mathscr{A}}}{\|g_{\mathrm{ref}}\|_{2,\mathscr{A}}}, \qquad \rho_g = \frac{\sum_{i\in\mathscr{A}}(g_{\mathrm{rec},i} - \bar{g}_{\mathrm{rec}})(g_{\mathrm{ref},i} - \bar{g}_{\mathrm{ref}})}{\sqrt{\sum_{i\in\mathscr{A}}(g_{\mathrm{rec},i} - \bar{g}_{\mathrm{rec}})^2 \sum_{i\in\mathscr{A}}(g_{\mathrm{ref},i} - \bar{g}_{\mathrm{ref}})^2}}.$$

Here $\|g\|_{2,\mathscr{A}} = (\sum_{i\in\mathscr{A}} g_i^2)^{1/2}$, where i indexes spatial samples. Uniform-grid quadrature factors cancel in relative errors. Suffixes rec and ref denote reconstruction and reference. Here the bars are means on the same set; ρ_g is the Pearson correlation. The budget masks in figure 5 and table 6 are the frozen sets $|S_{\mathrm{bud}}| \geq 0.02 \max|S_{\mathrm{bud}}|$

Table 6. Two-budget recovery of R_{xx}^{cl}. Errors are active-zone relative L^2 errors in percent. The heat-flux budget alone gives S, while the scalar-excess budget supplies Δ.

Mach	ε_Δ	$\varepsilon_R(\Delta=0)$	$\varepsilon_R(\Delta_{\mathrm{full}})$	$\varepsilon_R(\Delta_{\mathrm{bud}})$	$\rho_R(\Delta_{\mathrm{bud}})$
M2	3.757	64.362	3.356	4.427	0.99950
M3	3.457	63.225	3.101	3.735	0.99949
M5	2.182	63.112	2.219	2.544	0.99974

after plateau subtraction, containing 622, 536 and 725 points for Mach 2, 3 and 5. The masks from the archived and regenerated heat-budget channels coincide. The Shakhov channel check uses $|S| > 0.02\max|S|$ on its own plateau-corrected reference; the core-gap probe test uses $|\Delta| \geq 0.02\max|\Delta|$. The deposited DSMC postprocessor defines its default comparison mask as the union of points with $|\Delta| > 0.02\max|\Delta|$ or $|Q_\Delta| > 0.02\max|Q_\Delta|$, retaining cells with at least one logged collision (with a whole-domain fallback if fewer than ten points remain); its relative budget error uses measured collisional production as denominator and is quoted as a fraction rather than percent. The historical summary does not retain the mask arguments or its point count, so exact mask-command provenance for that diagnostic remains incomplete. The sensitivity and cross-method comparisons use the fixed aligned interval $|(x-x_s)/\lambda_1| \leq 15$, with the stated reference profile as denominator. Thus "active zone" denotes a specified comparison set, not one universal mask across experiments.

No direct R_{xx}^{cl} anchoring or supervised R data enter this reconstruction. Figure 5 shows the result. The first column verifies the scalar-excess budget. The middle column shows the rank-deficient heat-flux interpretation obtained by setting $\Delta = 0$, which gives R errors of about 63–64%. The third column shows the two-budget reconstruction. The errors drop to 4.43%, 3.73% and 2.54%, close to the reference error obtained when the full DVM Δ field is used. The correlations range from 0.99949 to 0.99974.

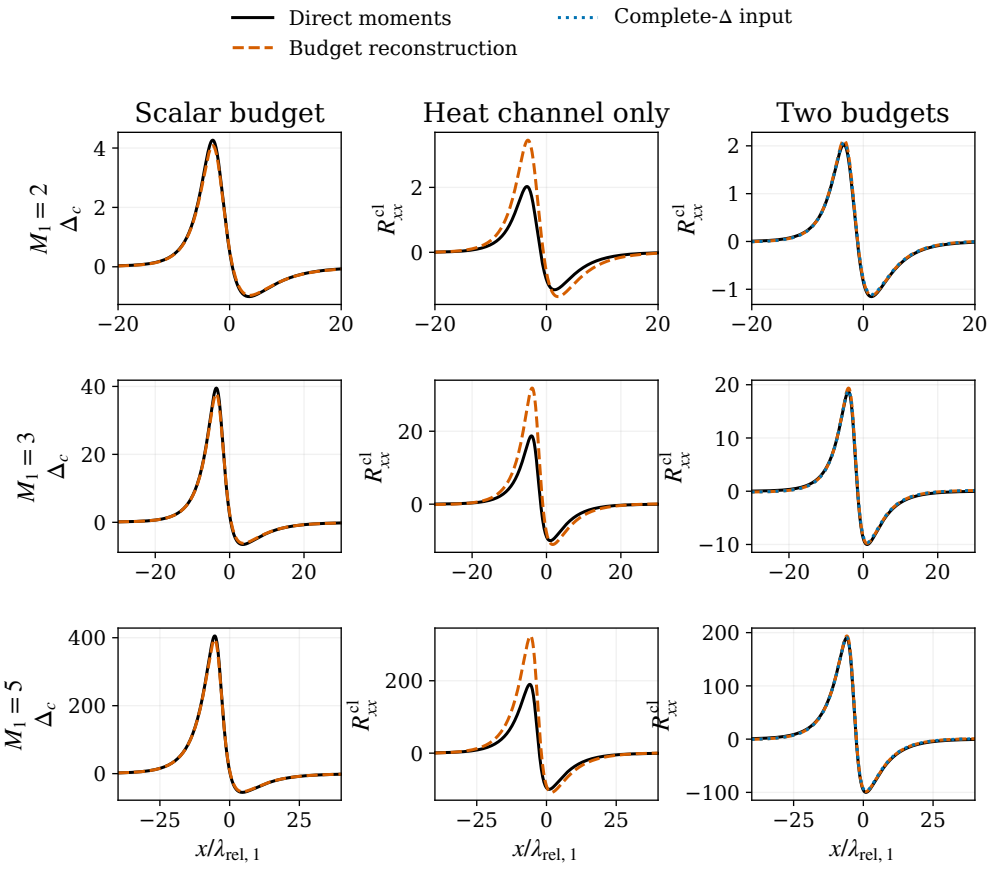


Figure 5. Two-budget recovery of the tensorial fourth-order closure moment. First column: scalar-excess budget reconstruction of Δ. Second column: heat-flux channel alone, where $R_{xx}^{\mathrm{cl}} \simeq S$ corresponds to setting $\Delta = 0$. Third column: two-budget reconstruction $R_{xx}^{\mathrm{cl}} \simeq S_{\mathrm{bud}} - \Delta_{\mathrm{bud}}/3$. The dotted curve shows the full-Δ reference reconstruction, i.e. the reference reconstruction obtained when the complete DVM scalar-excess field is available.

Table 7. Logged high-Mach BGK run parameters and upstream reservoir quadrature checks. Time steps are rounded as in the original logs; every run reaches the stated N_t. The final column is the stress of the fitted upstream discrete Maxwellian, normalised by $\rho_1 T_1 = 1$; its continuum equilibrium value is zero. It is a grid diagnostic, not physical upstream stress.

Case	Δt	N_t	$\sigma_{xx}[M_{d,1}]/(\rho_1 T_1)$
M6 full	2.2846×10^{-3}	120000	0.04233
M8 full	2.0097×10^{-3}	160000	0.1129
M12 full	1.3397×10^{-3}	220000	0.8413
M8 lightweight	2.3448×10^{-3}	100000	0.1639
M12 lightweight	1.5071×10^{-3}	140000	0.8257

5.4. *Fourth-order channels at higher Mach numbers*

The stored full BGK distributions extend the moment-level comparison to Mach 6, 8 and 12 (figure 6). These runs retain the constant-relaxation BGK model, $Pr = 1$ and $\tau = 1$, on the grids in table 3. Their spatial endpoints impose only incoming Maxwellian populations: $f(x_L, v) = M_{d,1}$ for $v_x > 0$ and $f(x_R, v) = M_{d,2}$ for $v_x < 0$, with the two reservoir states related by Rankine–Hugoniot conditions. Outgoing populations are determined by the interior solution. The stored distributions are single precision; their moments are re-evaluated in double precision, which reduces accumulation error but does not upgrade the precision of the underlying kinetic solution. Here Δ_d is measured relative to a Maxwellian matched on the discrete velocity grid, and $S_d = R^{\mathrm{cl}}_{xx} + \Delta_d/3$. No far-field plateau is subtracted. Increasing the matching iteration count from four to twelve changes Δ_d by at most 4.71×10^{-6} in relative L_2 norm across the five full/lightweight datasets.

The high-Mach integrations use first-order upwind streaming and the frozen-local-target BGK relaxation update, with $\Delta t = 0.75 \Delta x / v_{\max}$. Table 7 reports the logged time steps and completed step counts. Runs terminate at these prescribed counts, with periodic recentering and a 10^{-30} positivity floor, rather than a residual-based stationary stopping criterion. The different step counts and time steps also mean that full/lightweight comparisons do not isolate temporal error.

The upstream check exposes a limitation of these historical velocity grids. Matching density, mean velocity and scalar temperature does not enforce an isotropic pressure tensor. The narrow upstream thermal population remains at $T_1 = 1$ while the velocity box expands to cover the hotter downstream gas; the coarse transverse spacing then produces a nonzero equilibrium stress, reaching $0.8413 \rho_1 T_1$ in the full Mach-12 case. The similar defect in its lightweight partner shows why close agreement between two moment profiles can coexist with a shared quadrature error. Double-precision post-processing and discrete-equilibrium subtraction do not remove errors already introduced during kinetic evolution. The high-Mach profiles are therefore exploratory evidence about the stored moment channels, with equilibrium-quadrature and temporal limitations, rather than independently validated strong-shock solutions.

Within this scope, the higher-Mach profiles show that the scalar complement remains substantial in the stored stronger-shock distributions. The ratios $\max|\Delta_d/3| / \max|R^{\mathrm{cl}}_{xx}|$

are 0.714, 0.716 and 0.720 at Mach 6, 8 and 12, respectively. The positive upstream lobe and negative downstream lobe persist, while their amplitudes increase in the stored nondimensional units. Thus the additional evidence is persistence of a comparable scalar contribution at higher Mach number, rather than a new observation channel or an inferred universal Mach scaling. These peak ratios do not measure pointwise cancellation. In particular, the Mach-2–5 budget-recovery errors in table 6 are not extrapolated to these cases: the present extension evaluates moments directly from the distributions.

For Mach 8, full/lightweight relative L_2 differences in $(R_{xx}^{\mathrm{cl}}, \Delta_d, S_d)$ are $(0.274, 0.234, 0.253)\%$; at Mach 12 they are $(0.230, 0.272, 0.253)\%$. The comparison uses the common stored x/λ_1 interval, linear interpolation and the full-profile norm as denominator, without further alignment or plateau subtraction. Table 3 makes the differences between the paired configurations explicit. At Mach 8, the spatial and velocity point counts change at fixed domain extents. At Mach 12, both extents and point counts change; the full configuration even has a slightly larger Δx than the lightweight one. These comparisons therefore measure combined numerical sensitivity rather than isolated convergence. A separate matched-parameter Shakhov comparison with a published Mach-8 benchmark is given in appendix E.

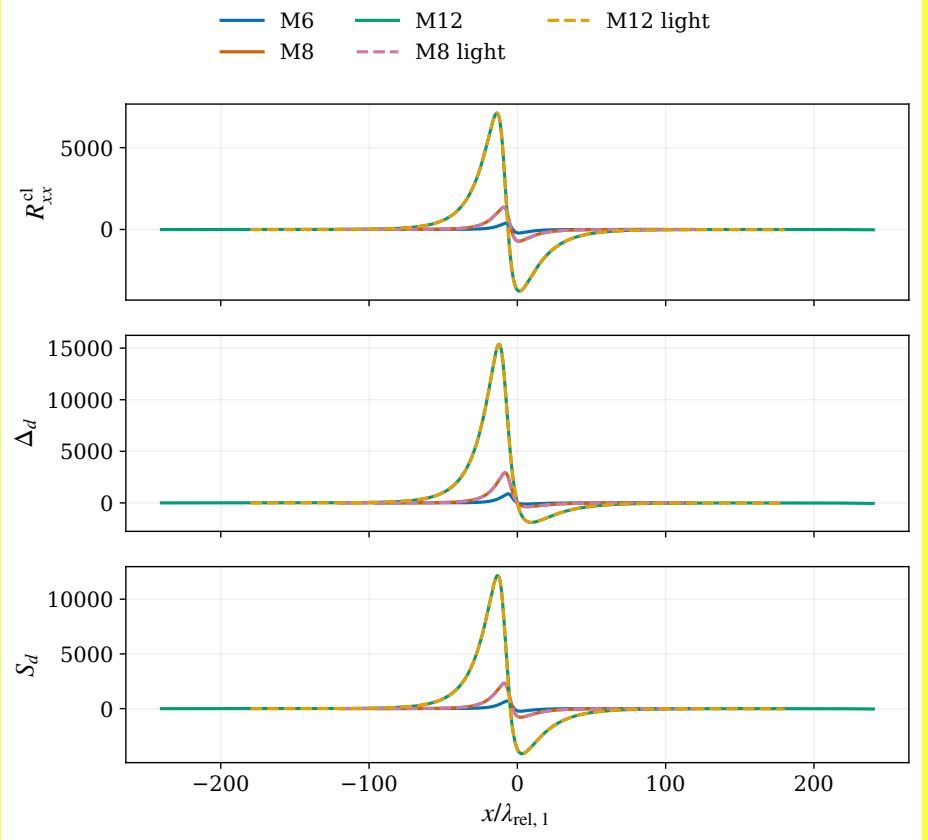


Figure 6. BGK fourth-order moments at Mach 6, 8 and 12 for the full configurations in table 3. Dashed curves show the lightweight Mach-8 and Mach-12 variants in the same table. The discrete-reference scalar excess is retained separately from the continuum-reference excess. The plotted fields use the stored upstream mean-free-path coordinate without plateau subtraction. Full/lightweight agreement measures combined numerical sensitivity and does not establish independent velocity-tail, spatial-grid or temporal convergence; the upstream equilibrium-quadrature defects are reported in table 7.

6. Sparse scalar-excess information

The budget reconstruction above uses full DVM moments to establish the physical channel. To test whether this channel represents reduced information rather than disguised direct access to R^{cl}_{xx}, we replace the full Δ field by a sparse scalar-excess probe. The DVM Δ profile is sampled at 24 shock-local interior points with two endpoint anchors and reconstructed with a smooth radial-basis-function (RBF) representation; no values of R^{cl}_{xx} are used. The tensorial closure is then computed from $R^{\mathrm{cl}}_{xx} = S - \Delta_{\mathrm{sparse}}/3$.

Figure 7 reports a separate interpolation test using its archived, fixed heat-budget channel. With 24 shock-local interior samples of Δ and two endpoint anchors, the tensorial errors are 4.33%, 4.49% and 3.77% for $M_1 = 2, 3, 5$; adding 1% probe noise keeps the error below 4.7% under that same contract (table 8). These results test scalar information reduction within the archived channel; they are not a new sparse-probe evaluation of the regenerated discrete-source budget in table 6. The full-field reference curves in this interpolation study belong to the same archived contract.

This is an interpolation and information-reduction experiment, not a theorem that 24 probes are sufficient in a universal or optimal sense.

The structural statement is instead that the missing information has the type of one scalar fourth-order field; the number of probes required to interpolate that field depends on smoothness, shock thickness, noise level and probe placement. The convergence curve in figure 7 should therefore be read as an operating study for the present shock family. Small numbers of probes already remove the order-unity ambiguity, and the 24-interior-probe case is a tested operating point within the fixed-channel interpolation experiment, not a certified sensor minimum or a comparison with the newly regenerated full-field budget. Thus the information needed to resolve the heat-flux ambiguity is not a dense R^{cl}_{xx} field; it is an independent scalar measure of isotropic fourth-order tail intensity. This distinction matters for experiments and reduced solvers: a scalar diagnostic, surrogate or auxiliary equation for tail intensity can close the heat-flux null space without supervising the whole tensorial closure field.

Table 8. Sparse scalar-excess reconstruction errors in the archived fixed-channel interpolation experiment of figure 7. All entries are ε_R in percent. The first three columns vary the noise in 24 interior Δ probes, supplemented by two endpoint anchors. The last three compare the heat-flux-only reconstruction ($\Delta = 0$), the full-Δ reference and the noise-free sparse reconstruction. Values are transcribed from the archived plot and rounded to two decimal places; this presentation does not supply missing raw-data provenance or constitute a new budget regeneration.

	Probe noise			Scalar information		
M_1	0%	0.5%	1%	$\Delta = 0$	Full Δ	Sparse Δ
2	4.33	4.42	4.52	64.36	3.36	4.33
3	4.49	4.59	4.68	63.23	3.10	4.49
5	3.77	3.91	4.04	63.11	2.22	3.77

Sensitivity to probe placement. The small errors above apply to the specified shock-local sampling layouts, not arbitrary sets of 24 points. Figure 8 and table 9 test this distinction on the completed Mach-5 baseline. With the same 24 interior probes and

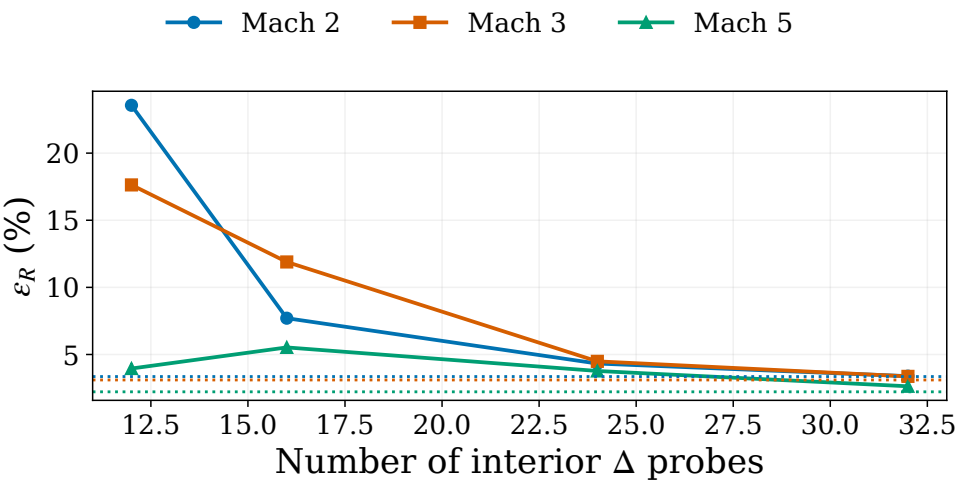


Figure 7. Convergence of the sparse scalar-excess reconstruction using the archived fixed heat-budget channel, separate from the regenerated budget in table 6. The error in R_{xx}^{cl} decreases as the number of shock-local interior Δ probes increases; two endpoint anchors supplement these probes. Dotted lines denote the full-Δ reference errors. The 24-interior-probe case is a tested operating point, not a certified sensor minimum. Its noise sensitivity and comparison with full-Δ and $\Delta = 0$ reconstructions are tabulated in table 8.

two boundary anchors, an active-uniform layout gives a 3.107% active-zone error in R_{xx}^{cl}, whereas a gap of width 25% of the active interval, centred at the largest scalar-excess magnitude, raises that error to 33.014%. The existing full-domain-uniform layout, with 24 total probes including endpoints, gives 17.385% and is not an exactly equal-count comparison. These controlled tests supply $S = R_{xx}^{\mathrm{cl}} + \Delta/3$ exactly to isolate scalar interpolation error; they do not revalidate the budget-derived channel or replace the earlier Mach-sweep errors. The active interval is chosen using the reference field, so the experiment also does not demonstrate blind sensor placement. A low leave-one-out error on observed probes does not guarantee accuracy within an unobserved core gap. The practical sparse-recovery claim therefore requires adequate core coverage.

Table 9. Mach-5 probe-placement sensitivity corresponding to figure 8. The final two columns are relative L_2 errors in percent on the active interval $|\Delta| \geq 0.02 \max|\Delta|$. Counts distinguish all probes from probes within that interval. The channel S is supplied exactly and no noise is added, so the test isolates scalar interpolation error. Full-domain uniform has fewer total probes and is not an equal-count comparison.

Layout	Total probes	Active probes	ε_Δ	ε_R
Active uniform	26	16	4.880	3.107
Core gap 25%	26	14	51.854	33.014
Full-domain uniform	24	7	27.305	17.385

7. Direct DSMC collision production of the scalar-excess source

The relaxation models above are useful because they isolate specific collision-model mechanisms. BGK and Shakhov relax the scalar excess as $-\Delta/Kn$, while ES-BGK adds a stress-quadratic target contribution. A stronger test is to measure the scalar-excess production from collisions without prescribing any relaxation target. We therefore perform a standalone DSMC calculation of a Mach-3 monatomic hard-sphere normal shock and

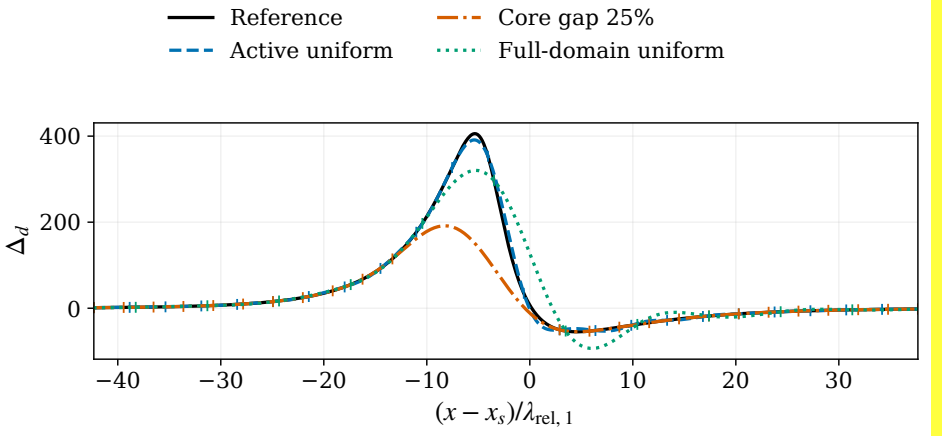


Figure 8. Controlled probe-placement sensitivity for the Mach-5 scalar channel. Reference and Gaussian RBF reconstructions, with probe locations marked. Relative L_2 errors over $|\Delta| \geq 0.02 \max|\Delta|$ are reported in table 9. Active-uniform and core-gap layouts have 26 total probes (16 and 14 in the active interval); full-domain uniform has 24 total (seven active). The active span is padded by 25% on either side; the gap width is 25% of the unpadded active span. The mean of the outermost 8% at each end is subtracted following the existing interpolation procedure. Gaussian-RBF parameters are selected from sparse data by leave-one-out validation: 28/32/48 centres, width multipliers 0.5/0.6/0.8/1/1.2/1.5 and ridge parameters $10^{-10}/10^{-8}/10^{-6}/10^{-4}/10^{-2}$. No noise is added; S is exact. This is a controlled interpolation stress test, not full two-budget validation.

accumulate the production of the central fourth scalar moment directly during accepted binary collisions.

For each accepted collision pair in cell j, let c_1 and c_2 denote peculiar velocities relative to the instantaneous cell mean velocity before collision, and let primed quantities denote the post-collision velocities after elastic hard-sphere scattering. The cell-wise scalar-excess production is sampled as

$$Q^{\mathrm{DSMC}}_{\Delta,j} = \frac{1}{V_j \Delta t_{\mathrm{samp}}} \sum_{\mathrm{coll}\in \mathrm{j}} \left[|c_1'|^4 + |c_2'|^4 - |c_1|^4 - |c_2|^4 \right] w_p, \tag{7.1}$$

Here j labels a collision cell of volume V_j, and the sum runs over accepted binary collisions accumulated during the exposure Δt_{samp}. The constant weight w_p is the mass represented by one simulation particle (equivalently molecular count in unit molecular-mass units). Pair labels 1 and 2 denote the two particles, using the same pre-collision cell mean u_j: $c_\ell = v_\ell - u_j$ and $c'_\ell = v'_\ell - u_j$. Primes here denote post-collision values, whereas u'_x denotes a spatial derivative. The exposure is the time represented by accumulated collision events, rather than the spacing between stored moment samples. $Q(f,f)$ is the Boltzmann binary-collision operator. This quantity is the DSMC counterpart of $\int |c|^4 Q(f,f)\,\mathrm{d}v$. It is not imposed to be proportional to $-\Delta$; it is a direct measurement of how binary collisions create or destroy the scalar-excess moment in the shock layer.

For this hard-sphere campaign, $\lambda_1 = (\sqrt{2} n_1 \Sigma_T)^{-1}$ is the upstream mean free path, where n_1 is molecular number density and Σ_T the total hard-sphere cross-section. PPC denotes the nominal upstream simulation-particle count per spatial cell; independent seeds denote independent pseudorandom initialisations. The DSMC run uses 400 spatial cells over $[-60, 60]\lambda_1$, upstream particles per cell equal to 120, time step 0.005, and eight

independent seeds. Sampling starts after 70000 steps and continues to 180000 steps, with sampling every 10 steps. The realisations are aligned by their density-midpoint shock locations before averaging. Each realisation has 11001 stored moment samples and an average of 8.2×10^5 accepted collisions per active-zone cell. This alignment is essential: the scalar-excess budget contains the derivative of the fifth-order flux J_Δ, and direct averaging of unaligned shock profiles smears the derivative even when the lower-order shock structure appears converged.

Figure 9 shows the result. Panel (a) verifies that the direct DSMC calculation samples the same nonequilibrium shock layer in which the heat flux and normal stress are concentrated. Panel (b) shows the scalar excess Δ. Panel (c) compares the directly accumulated collisional production Q_Δ^{DSMC} with the spatial scalar-excess budget,

$$Q_\Delta^{\mathrm{budget}} = \partial_x J_\Delta + 4(\partial_x u_x) K_\Delta . \tag{7.2}$$

The derivative is evaluated using a local cubic polynomial with a seven-point stencil; windows of 5–9 points give similar results, while larger windows over-smooth the narrow shock layer. Specifically, stencil windows of 5, 7 and 9 give active-zone errors 0.173, 0.170 and 0.188, and correlations 0.986, 0.987 and 0.984, respectively. The active-zone comparison reported in figure 9 gives a relative error of 0.170 and a correlation of 0.987. For DSMC, both sides of (7.2) are difficult quantities: one is a noisy collision-production sample and the other contains a derivative of a fifth-order flux. The comparison supports the collision-production interpretation of the scalar-excess channel within the tested implementation; the sampler audit below limits its use as a validation of the Boltzmann operator.

Panel (d) compares the normalised collision production with the normalised $-\Delta$ profile. The correlation is 0.968, showing that the DSMC collisions relax the scalar excess in the shock core with the same qualitative sign structure as BGK and Shakhov. The comparison is deliberately normalised: the DSMC production is not assumed to follow the BGK/Shakhov unit relaxation rate, and the effective proportionality varies through the shock. This is precisely the point. The scalar channel identified by the two-budget theory is not a relaxation-model artefact; it is a collisional channel whose source law becomes model-dependent once one moves from BGK to Shakhov, ES-BGK, ES-FP and DSMC/Boltzmann physics.

The DSMC calculation extends the source diagnostics to directly sampled binary collisions. Together with the model sources in table 5, it illustrates how a common heat-flux observation channel can coexist with different scalar-production laws. The pair-selection limitation discussed in the sampler audit below applies to the quantitative DSMC comparisons.

Particle-number audit and interpretation. The original ensemble with 120 particles per cell (PPC120) is supplemented by PPC60, 240, 480, 960, 1920 and 3840, each with eight seeds and the same grid, time step and sampling schedule, giving 56 runs. Figure 10 shows the high-PPC moment profiles; appendix B reports the directly logged collision-source audit. Between PPC1920 and PPC3840, aligned ensemble-mean R_{xx}^{cl}, Δ and S differences are approximately 0.385%, 0.391% and 0.309%, and the collision-

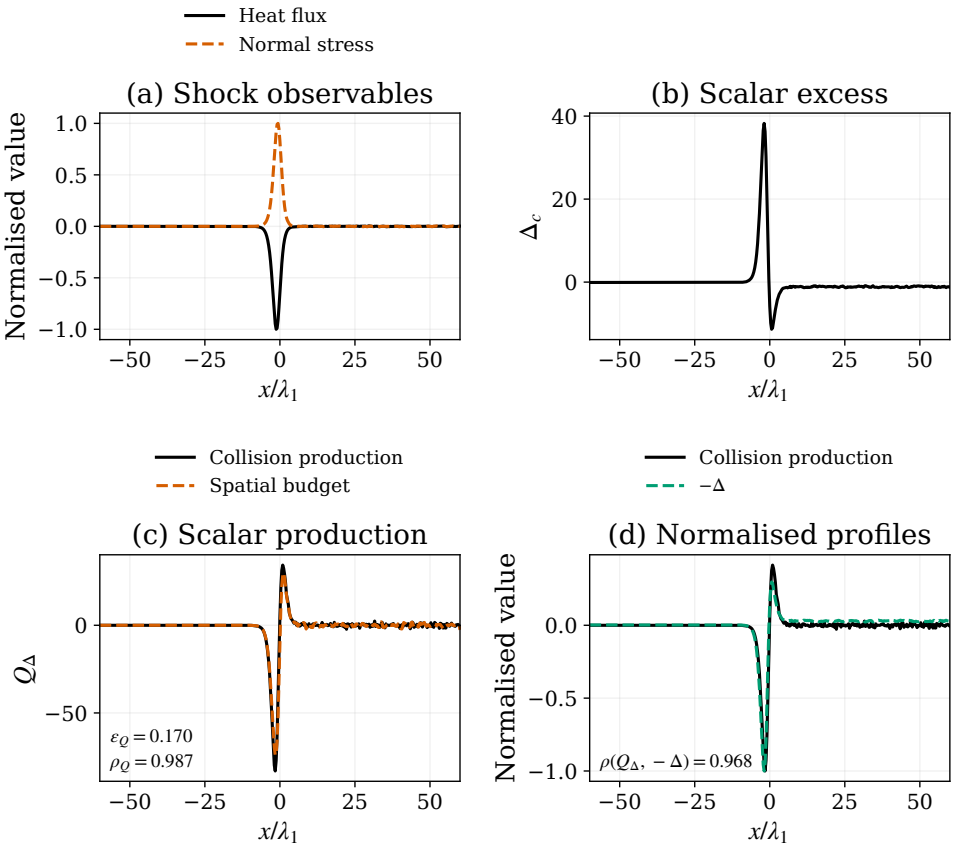


Figure 9. Direct DSMC scalar-excess production in a Mach-3 monatomic hard-sphere shock. (a) Normalised heat flux and normal stress identify the nonequilibrium shock layer. (b) Scalar fourth-order excess Δ. (c) Direct collisional production Q_Δ^{DSMC}, accumulated from pre- and post-collision changes of $|c|^4$, compared with the spatial budget $\partial_x J_\Delta + 4u_x^{\Gamma} K_\Delta$. Realisations are aligned at their density midpoints before averaging; the plotted x/λ_1 is this aligned coordinate, $(x - x_s)/\lambda_1$. The active-zone relative error is 0.170 and the correlation is 0.987. (d) Normalised Q_Δ^{DSMC} and $-\Delta$ have correlation 0.968, indicating strong relaxation-like alignment without assuming a BGK relaxation rate.

source difference is 0.668%. These finite-sample differences do not establish statistical equivalence or independence from the sampling window.
Moreover, an audit identified a pair-selection bias in the original DSMC sampler; corrected-sampler pilots are separate and have not been mixed into this ensemble. Consequently, the original-source budget correlations remain diagnostics of that implementation, not a bias-free validation of the Boltzmann collision operator. Between-seed intervals do not account for this systematic error.

8. Consequences and outlook for closure modelling

The identifiability result has a direct consequence for closure modelling. If a model is trained, fitted or regularised only through the heat-flux budget, then all states related by

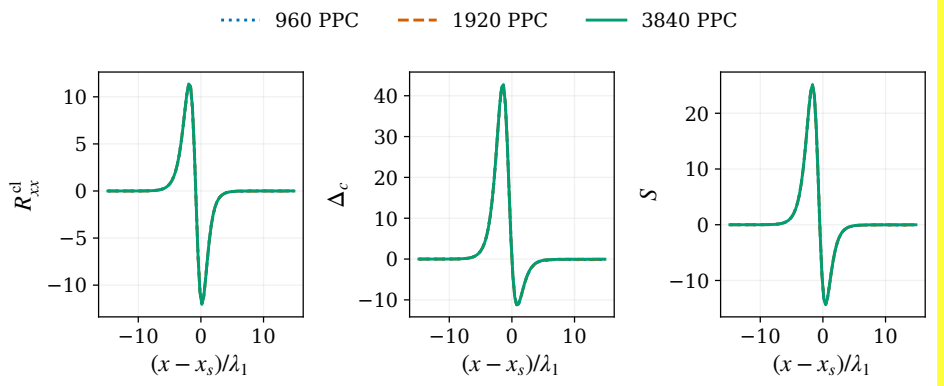


Figure 10. Mach-3 hard-sphere DSMC high-order moments at PPC960, 1920 and 3840. Lines show eight-seed means and bands pointwise 95% Student-t intervals with seven degrees of freedom. Each time-averaged profile is aligned at its $n = 2$ crossing by linear interpolation, without smoothing; $\lambda_1 = 1$ in the DSMC units. The bands exclude systematic sampler error and are not simultaneous confidence bands. These are original-sampler results; the identified pair-selection bias remains unresolved. Alignment does not remove temporal shock wandering. The Mach-3 DSMC fields are not a direct same-case validation of the Mach-5 DVM comparisons.

(3.3) are observationally equivalent to that budget. A one-parameter family

$$R^{cl}_{xx,\eta} = R^{cl}_{xx} + \eta(x), \qquad \Delta_\eta = \Delta - 3\eta(x) \tag{8.1}$$

produces exactly the same observed channel S for any shock-local field $\eta(x)$. Two closures can therefore agree in the heat-flux residual and disagree in the tensorial closure moment required by an R26-type model. This is not merely a diagnostic inconvenience. The tensorial and scalar parts enter different members of a moment hierarchy and represent different pieces of the nonequilibrium distribution: traceless anisotropy and isotropic tail intensity. Treating their sum as the tensorial moment itself can therefore produce a closure with the correct observed energy-transport channel and the wrong internal split. In R26 language, such a closure would appear successful when inspected through the heat-flux equation, but it would supply the wrong tensorial fourth-order state to any equation, wall condition or regularisation term that depends on R^{cl}_{ij} rather than on the composite $A_{ij} = R^{cl}_{ij} + \Delta\,\delta_{ij}/3$.

These shocks provide a reference test for reduced kinetic solvers, neural closures and moment models: recover S, supply independent scalar-excess information, and then infer R^{cl}_{xx}. Agreement in low-order profiles alone cannot test this split. The sparse-probe results show that the scalar information can be sampled at a limited number of locations when those locations adequately cover the shock core.

The ES-BGK and ES-FP diagnostics (table 4 and figure 3) modify scalar production through stress anisotropy while retaining the heat-flux observation channel. Independently evolved ES-BGK Mach-2/3 fields are reported in appendix G; these finite-time calculations preserve positive states but have not passed the specified stationarity criterion. On the same DVM profiles, ES-FP gives the larger correction.

Figure 9 adds the binary-collision diagnostic, subject to the original-sampler bias in §7. Thus testing the observation map and testing the collision-source law remain separate requirements.

In multidimensional flows, the observed heat-flux quantity is the divergence of the composite tensor $R^{\mathrm{cl}}_{ij} + \Delta\,\delta_{ij}/3$, so additional tensorial information is needed to reconstruct the full closure tensor. The one-dimensional shock studied here is therefore the clean base problem, not the end point of the theory.

Relation to realizability and closed-system stability. The family in the displayed null-space family describes an unobserved direction in one balance law, not a family of admissible solutions of a closed R26 system. A residual that observes only S does not penalise changes along that direction or test their moment realizability. The scalar complement removes this observational freedom; positivity and realizability constraints must still be imposed independently. Hyperbolicity additionally depends on the constitutive flux laws for m_{ijk}, R^{cl}_{ij} and Δ. No closed-system principal symbol is evaluated here, so removing the null direction does not prove a real spectrum, removal of the Bobylev instability or hypoellipticity. In multiple dimensions, complementary observations would need to resolve the irreducible rotational components, for example in a representation of the Lie algebra $\mathfrak{so}(3)$ of three-dimensional rotations, rather than append only one scalar observation.
The two-budget procedure is primarily an identifiability diagnostic. It is constructive when an invertible scalar source law, the required fluxes and an anchor are supplied, as for these BGK references. For general Boltzmann production, a measured source does not alone determine Δ algebraically.
Scope of the channel result. Invertibility of the two-observation map does not imply positivity of an underlying distribution, hyperbolicity of a closed moment system, or an entropy inequality. No closed flux Jacobian is constructed here, and no restoration of a real spectrum or hypoellipticity is claimed. In multiple dimensions, observing a tensor divergence does not identify all tensor components: adding a scalar channel is not generally sufficient. The present reservoir-boundary shock calculation supplies no solid-wall moment boundary law. Polyatomic internal energies would enlarge the moment state and introduce additional exchange channels; the one-dimensional monatomic rank calculation cannot be transferred unchanged. Finally, this work does not train a constitutive closure or establish its Chapman–Enskog limit. Varying a mean-free-path coordinate alone is not an independent macroscopic hydrodynamic-limit test. The reported 2.54–4.43% recovery range concerns the tested BGK references, not arbitrary molecular cross-sections.

9. Conclusions

The heat-flux budget of a monatomic kinetic normal shock constrains a combined fourth-order energy-transport channel, not the tensorial closure moment alone. In one dimension this channel is $S = R^{\mathrm{cl}}_{xx} + \Delta/3$, and the observation map from $(R^{\mathrm{cl}}_{xx}, \Delta)$ to S has a one-dimensional null space. This projection property explains why residual agreement in the heat-flux equation cannot by itself certify recovery of every R26-level closure variable.

The missing scalar channel can be supplied constructively. A DVM-consistent scalar-excess budget supplies the missing scalar field, and the two budgets recover R_{xx}^{cl} without direct R data. For the tested $Pr = 1$ BGK shocks at Mach 2, 3 and 5, the error falls from about 63–64% to 2.54–4.43%. With adequate shock-core coverage, sparse scalar-excess probes approach the full-field recovery accuracy in the tested shocks. The missing information is one scalar channel; the number and placement of probes needed to reconstruct it depend on the shock profile. These errors do not include a Mach-4 budget test or the separate $Pr = 2/3$ cross-method campaign.

The collision-model analysis shows that the observation channel and the source law must be kept separate. Table 5 summarises the ladder. Shakhov corrects the heat-flux relaxation to $Pr = 2/3$ while leaving the even scalar-excess source invariant. ES-BGK leaves the heat-flux observation channel unchanged but modifies the scalar-excess source through a stress-quadratic anisotropic-Gaussian target, contributing about 4.3–4.5% of $\|\Delta\|$ in the present Mach 2–5 profiles. ES-FP gives a larger anisotropic-diffusion correction of about 21.5–22.4% while remaining strongly correlated with the BGK/Shakhov source. Finally, DSMC measures the corresponding fourth-order production directly from accepted binary collisions. In the original-sampler Mach-3 DSMC ensemble, the direct collisional production and spatial scalar-excess budget have correlation 0.987, while the production and $-\Delta$ have correlation 0.968. The identified pair-selection bias limits both comparisons to diagnostics of that implementation. Thus the scalar-excess channel is not an artefact of BGK-type relaxation targets; it is a physical collision-production channel whose specific source law changes as one moves from BGK and Shakhov to ES-BGK, ES-FP and Boltzmann/DSMC.

The higher-Mach BGK distributions extend the direct moment comparison to Mach 6, 8 and 12 on the case-dependent spatial and velocity domains in table 3. Their peak scalar-to-tensorial ratios $\max|\Delta_d/3|/\max|R_{xx}^{\mathrm{cl}}|$ are 0.714, 0.716 and 0.720, respectively. The scalar complement remains substantial in these stored distributions, subject to the equilibrium-quadrature defects in table 7. Full/lightweight profile differences of roughly 0.23–0.27% at Mach 8 and 12 measure combined numerical sensitivity; they do not extend the Mach-2–5 budget-recovery result or establish isolated convergence. The separate Mach-8 Shakhov comparison in appendix E tests density and temperature against published profiles, with stationarity and positivity limitations explicitly retained.

The numerical evidence must be distinguished from the identifiability statement. The observation-map null space is an algebraic property; small discretisation sensitivities do not prove hyperbolicity, entropy stability or convergence of every collision model. The sparse-recovery operating point requires shock-core coverage, and the corrected-sampler comparison quantifies rather than eliminates uncertainty associated with pair selection. The final velocity-grid comparison is complete. At PPC 960, the eight-seed differences in the three moment profiles are below 1%, while the collision-source difference is 1.413%; these differences are comparable to the estimated sampling uncertainty. Neither this comparison nor the entropy diagnostic establishes universal model validation (appendix C).

As a separate profile-level benchmark, appendix D compares high-particle-count Bird DSMC shocks at Mach 2–5 with available Shakhov DVM snapshots and corrected-Shakhov R26 profiles. This comparison complements, but does not replace, the DSMC collision-production and sampler-sensitivity studies: the new runs contain one seed per Mach and do not record the collision-wise Q_Δ observable used in §7.

Funding

This research received no specific grant from any funding agency, commercial or not-for-profit sectors.

Declaration of AI-assisted tools

ChatGPT (OpenAI) and Claude (Anthropic) assisted with language editing, numerical-script debugging and critical review of the manuscript and response letters. The author is responsible for verifying the numerical results, figures, equations, references and scientific conclusions; AI suggestions were not treated as independent scientific evidence.

Declaration of interests

The author reports no conflict of interest.

Data availability statement

Manuscript sources, figure and table files, recoverable solver and analysis source snapshots, compact numerical profiles and run/check records are deposited in the public repository `https://github.com/Ehsan-Roohi/DVM-JFM`; the scientific deposit is identified by commit `4c3d311`, and the accompanying reader-facing revision is identified by tag `reviewer-revision-20261010` (Roohi 2026*b*). The independent ES-BGK Mach-2/3 deposit, including all six completed $t = 400$ reduced-distribution checkpoints, budget arrays, profiles and reproducible comparison scripts, is identified separately by tag `esbgk-m2-m3-t400-20261011`. Its own deposit manifest records file sizes and SHA-256 hashes. Continuations are kept outside this frozen finite-time deposit. The original publication manifest records file sizes and SHA-256 hashes, while its solver-specification and coverage audits distinguish the collision-model campaigns, historical and corrected calculations, and unresolved provenance gaps. The bundled Mach-2 and Mach-3 raw-profile comparison scripts reproduce the deposited comparison tables. This deposit does not contain all multi-gigabyte velocity-distribution/restart files, and some archived drivers require adaptation of machine-specific paths. The Bird DSMC original remains an external dependency; the repository supplies the documented modifications and original-source identification. The archived budgets have been regenerated with their documented discrete-source contracts. The full distribution used for Figure 4 is included in the public repository with its hash; a complete public deposit of the larger BGK raw distributions remains outstanding. No DOI has yet been assigned.

Appendix A. DVM numerical sensitivity

This appendix concerns the Mach-5 BGK sensitivity campaign. The velocity integrals use uniform tensor-product composite-trapezoidal weights, including half endpoint weights on each axis. Streaming is advanced explicitly with first-order upwind differences and $\Delta t = C_{\mathrm{CFL}}\Delta x/v_{\max}$. A local discrete Maxwellian is formed by iteratively matching density, streamwise velocity and temperature, followed by the frozen-local-target update $f^{n+1} = M + (f^{*} - M)\exp(-\Delta t)$.

The code uses unit relaxation rate in these computational coordinates; a macroscopic-length definition of *Kn* must not be inserted into these arrays without transforming the transport and source scales together. Maxwellian reservoirs supply incoming velocities at the left and right boundaries, at Rankine–Hugoniot states. There are no solid walls.

The implementation recentres the shock periodically and applies a positivity floor of 10^{-30}. These operations are part of the numerical method, not proofs of moment-budget preservation. Runs end at the prescribed step counts; the reported late-profile changes diagnose stationarity but are not an implemented residual-based stopping test. Conservation of low-order moments by the fitted Maxwellian does not guarantee exact equilibrium values of higher moments on a finite velocity grid. For each field $g \in \{\rho, T, q_x, R^{\mathrm{cl}}_{xx}, \Delta_d, S\}$, the comparison uses $100\|g_{\mathrm{base}} - g_{\mathrm{test}}\|_2/\|g_{\mathrm{test}}\|_2$ on 601 equally spaced points in $|(x - x_s)/\lambda_1| \le 15$. Each profile is independently aligned at its density midpoint and linearly interpolated without additional smoothing or plateau correction. The test profile is the denominator, not an exact reference solution.

The baseline has $N_x = 2200$, half-domain $80\lambda_1$, velocity grid $141 \times 27 \times 27$ and $v_{\max} = 22$. The enlarged velocity domain uses $169 \times 33 \times 33$ points with longitudinal bound 26.4 and transverse bound 27.076923. Velocity refinement uses $211 \times 40 \times 40$ points at the baseline extent. Spatial refinement uses 3299 points; the enlarged spatial half-domain is $120\lambda_1$ with 3300 points. The time-step test halves the baseline CFL coefficient from 0.75 to 0.375. Prescribed step counts for baseline, velocity extent, velocity refinement, spatial refinement, half time step and spatial extent are 52000, 62400, 52000, 78000, 104000 and 52000, respectively. All six integrations reached their prescribed end steps; final profiles, matching checkpoints and archive checksums were verified.

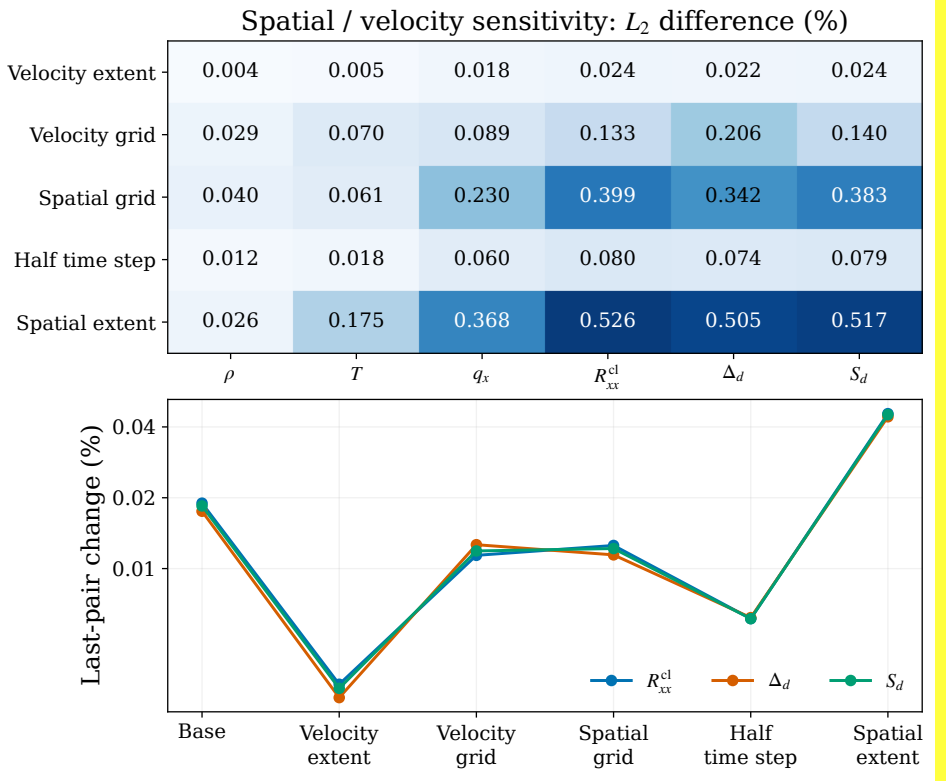


Figure 11. Mach-5 BGK DVM sensitivity summary. Left: relative L_2 differences defined in the text. Right: relative changes between the last two saved profiles, at step pairs 50000–52000, 62000–62400, 50000–52000, 76000–78000, 102000–104000 and 50000–52000 in the displayed case order. Unequal intervals are not comparable time derivatives. Two levels do not establish asymptotic convergence order; spatial refinement also changes the time step. These diagnostics introduce no new stopping threshold.

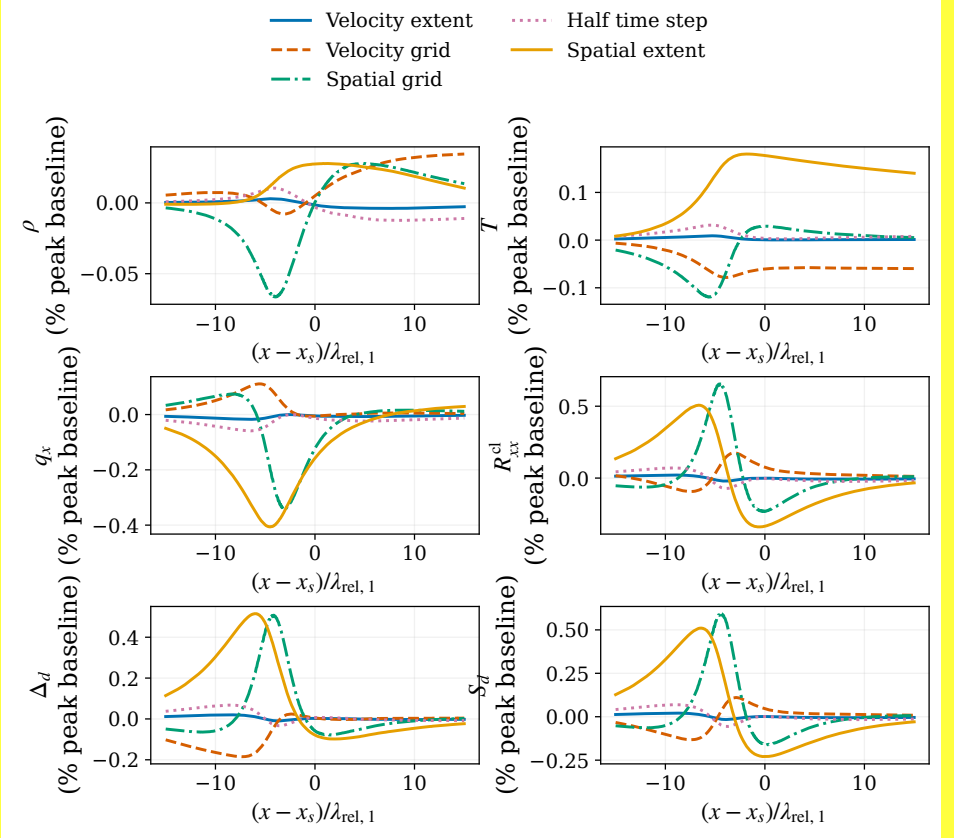


Figure 12. Aligned DVM profile differences for the same tests as figure 11. Each ordinate is $100(g_{test} - g_{base})/\max|g_{base}|$, with the maximum taken over the plotted interval. Peak normalisation avoids division by zero-crossing values. Δ_d uses the discrete-Maxwellian reference and is not the raw DSMC fourth-moment excess. This comparison does not directly measure omitted velocity-tail integrals or resolve distribution bimodality.

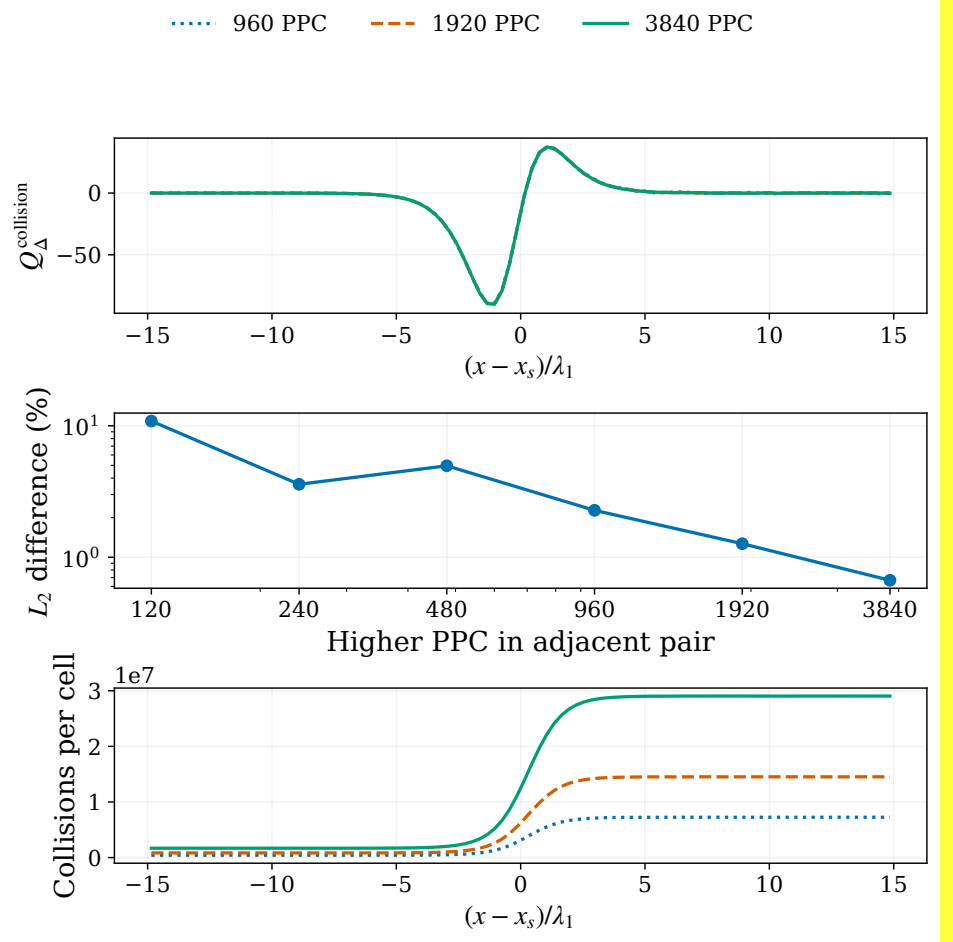


Figure 13. Directly logged scalar collision-source sensitivity. (a) Eight-seed mean and pointwise 95% Student-t intervals for the three highest PPC levels. (b) Adjacent-PPC relative L_2 differences of aligned mean source profiles, normalised by the higher-PPC profile over $|(x - x_s)/\lambda_1| \leq 15$. (c) Mean accumulated accepted-collision counts per cell, interpolated to the same aligned coordinate. No spatial smoothing is applied. Systematic pair-selection error is not included in the uncertainty bands.

Appendix B. DSMC particle-number sensitivity

The seven PPC levels retain 400 spatial cells, time step 0.005 and 180000 steps, with sampling after step 70000 every tenth step. All eight seeds at each level use the original sampler. Figure 13 holds this sampling schedule fixed and therefore does not establish averaging-window independence or reconfirm the historical collision-source/spatial-budget correlation of 0.987. Adjacent-PPC source differences are 10.870%, 3.589%, 4.963%, 2.275%, 1.268% and 0.668% for the successive pairs from 60–120 through 1920–3840. The trend need not be monotone because finite-seed noise and systematic effects remain. Collision counts are accumulated event counts, not independent sample counts or effective sample sizes.

Appendix C. Additional statistical and operator diagnostics

Corrected pair selection. A separate ensemble uses the corrected pair-selection implementation at PPC 120 and 960, with eight seeds per level and otherwise the same 400-cell grid, time step 0.005, 180000 steps and sampling schedule. Original and corrected checkpoints are version-specific. The original controls are reused; the corrected results are not substituted into the original PPC sequence. Each seed is aligned at the density midpoint before linear interpolation over $|(x - x_s)/\lambda_1| \leq 15$. A bar denotes the mean over eight seed profiles. Suffixes c and o identify corrected and original samplers; $s(g_c - g_o)$ is the pointwise sample standard deviation of their eight matched-label differences. For an observable g, the reported difference is $100\|\bar{g}_c - \bar{g}_o\|_2/\|\bar{g}_o\|_2$. A descriptive uncertainty scale is $100\|s(g_c - g_o)/\sqrt{8}\|_2/\|\bar{g}_o\|_2$, using matched seed labels. Random-number trajectories diverge after the algorithm change, so matching labels does not guarantee variance reduction. The differences and their descriptive uncertainty scales are reported in table 10.

Table 10. Eight-seed corrected-versus-original DSMC sampler comparison. Entries are percentages. Differences use the original ensemble mean as the denominator; the descriptive uncertainty scale is the norm of the paired-label standard error, as defined in appendix C. Matching seed labels does not guarantee variance reduction. These values are not significance or equivalence tests.

PPC	Measure	R_{xx}^{cl}	Δ	S	Q_Δ
120	Difference	3.611	2.851	3.288	3.524
	Uncertainty scale	3.304	2.497	2.869	3.859
960	Difference	0.933	0.734	0.800	1.413
	Uncertainty scale	0.971	0.718	0.809	1.296

Thus the sub-percent moment differences at PPC 960 are comparable to sampling uncertainty. This is not an equivalence test, a proof that the correction has no effect, or a convergence study of the corrected sampler.

Sampling-window diagnostic. For the eight original PPC-3840 seeds, ten disjoint 10000-step blocks are reconstructed over steps 80000–180000. Moment sums are differenced using sample counts; collision-source sums use their accumulated sampling-time exposure. Each block is independently aligned. Differences between the ensemble means of the first and last five blocks are 0.285%, 0.262%, 0.193% and 1.034% for $R_{xx}^{\mathrm{cl}}, \Delta, S, Q_\Delta$, respectively. Ten blocks do not establish full independence or a reliable effective sample size. Percentile intervals of a nonnegative norm cannot be interpreted as tests of zero drift.

Velocity tails and entropy. Twenty selected spatial rows of each Mach-5 baseline, enlarged-velocity-domain and refined-velocity-grid final distribution were inspected. These sampled tail diagnostics do not cover the Mach-6, Mach-8 or Mach-12 distributions. The largest resolved $|c|^4 f$ fraction in the outer ten percent of the velocity box was approximately 3×10^{-8} for the baseline/refined grids and 4.4×10^{-12} for the enlarged box. These are resolved outer-box contributions, not certified bounds on omitted tails beyond the computational domain.

The local ES-FP operator was evaluated on these BGK reference distributions using the full temperature tensor, $\beta = -5/4$ and unit collision rate. On the twenty inspected rows of each Mach-5 case, the minimum eigenvalue of $D/(2\nu T)$ was approximately 0.11160 for the baseline, 0.11157 for the enlarged velocity domain and 0.11239 for the refined velocity grid. Diffusion was therefore positive definite at these sampled states. This selected-row check does not establish admissibility at every spatial point, for the legacy Mach-2/3 profiles, or during an ES-FP evolution. Second- and fourth-order derivatives of $\log f$, boundary entropy fluxes, integration-by-parts defects and same-grid continuum-Maxwellian controls were compared. Computed shock-core entropy production was positive, but derivative-order sensitivity was 15.93% on the baseline grid and 9.17% on the refined grid. The maximum core energy-source residual, normalised by ρT, decreased from 0.103 to 0.01995 in these rate units. Negative upstream values on the coarse grid also occurred for its Maxwellian control; they cannot be identified as physical entropy violations. Conversely, these diagnostics do not establish an H-theorem or an entropy-stable ES-FP shock solution.

Shakhov clipping budget. In the separate 20-sweep diagnostic, maximum single-event clipping increments normalised by ρ, ρT and ρT^2 were 4.695×10^{-5} for mass, 1.327×10^{-3} for energy and 7.889×10^{-3} for the fourth scalar moment. These maxima are not accumulated errors at a fixed cell or steady-state error estimates. Clipping is therefore not assumed to preserve the analytic source-neutrality property of the unmodified Shakhov correction.

151 ## Appendix D. High-particle DSMC profiles and cross-method context

The Shakhov DVM curves in this appendix are a separate $Pr = 2/3$ campaign, not the $Pr = 1$ BGK budget references. The reduced distributions are $g = \int f \, dv_y dv_z$, $h = \int (v_y^2 + v_z^2) f \, dv_y dv_z$ and $k = \int (v_y^2 + v_z^2)^2 f \, dv_y dv_z$; retaining k permits the axial fourth-order quantities to be evaluated. Uniform cell-centred spatial grids and longitudinal composite-trapezoidal quadrature are used. Streaming is first-order explicit upwind with $\Delta t = 0.8 \Delta x / v_{\max}$. The local collision update uses the exponential relaxation factors $\exp(-\Delta t/\tau)$ and $\exp(-Pr \Delta t/\tau)$, with $\mu = T$, $\tau = 1/\rho$, and no distribution clipping. Incoming populations are supplied by the upstream and downstream Rankine–Hugoniot Maxwellian reservoirs; outgoing populations come from the interior. The plotted Mach-2 profile has 1280 cells on $[-30, 30]$ and 321 longitudinal velocities on $[-16, 16]$. The Mach-3, Mach-4 and Mach-5 profiles use 1920, 2560 and 3200 cells on $[-45, 45]$, $[-60, 60]$ and $[-75, 75]$, respectively, with 801 velocities on $[-40, 40]$. Their saved times are 238.125, 173.4375 and 143.4375. These are the archived plotted snapshots, not a claim that subsequently evolving jobs remained at those times.

For this campaign, stationarity requires five consecutive checks with the maximum fieldwise relative change below 10^{-7} over all eight saved fields $(\rho, u_x, T, q_x, \sigma_{xx}, R_{xx}^{\mathrm{cl}}, \Delta, S)$, with each denominator bounded below by 10^{-8}. Checks are separated by approximately two computational time units. The plotted Mach-3–5 receipts have zero qualifying consecutive checks. This criterion differs from that of the separate Mach-8 benchmark in appendix E.

The corrected-Shakhov R26 comparison uses the same $Pr = 2/3$ and $\mu = T$, $\tau = 1/\rho$ transport parameters, but solves a steady moment boundary-value problem. For the archived Mach-3 result, a piecewise degree-four representation on 4012 elements gives 16049 native points over $[-45, 45]$, with an eight-point Gauss residual objective and a sparse augmented Levenberg–Marquardt solve. Independent qualification evaluates the full corrected tensor PDE on the native points, finite profiles and positive pressure, with scaled PDE and boundary/phase tolerances 0.01 and 10^{-4}. The measured native PDE residual is 0.0046433 and the boundary/phase residual 1.51×10^{-11}. The runner stopped at its 30-iteration limit with automatic acceptance disabled; the separate independent checker subsequently passed the stated numerical criteria. Thus "accepted" here denotes this finite-tolerance independent qualification, not a zero residual, uniqueness, or agreement with a kinetic collision operator. The Mach-2 corrected profile has 3841 native points. Only these two corrected R26 profiles enter the present cross-method comparison; no corrected Mach-4 or Mach-5 R26 solution is included.

We add a separate Bird DSMC profile set at $M_1 = 2, 3, 4, 5$, with 2000 upstream particles per cell, 800 spatial cells, $\Delta t = 0.005$, and sampling through $t = 400$ after a warm-up to $t = 200$ (Bird 1994). The collision model is VHS with $\omega = 1$ and inverse scattering parameter $\alpha_{\mathrm{VSS}} = 1$ in the variable soft sphere (VSS) notation (isotropic scattering); ω is the viscosity exponent and starred quantities are divided by their corresponding upstream references; the transport law is $\mu^* = T^*$ and $\tau^* = 1/\rho^*$. Each Mach number has one independent seed, 800 total blocks, and 400 sampled blocks after warm-up. The high-particle runs use the original Bird sampler, so the pair-selection limitation described in appendix C remains applicable. For the Mach-2 quantitative comparison, profiles are aligned by a density-midpoint translation only and evaluated on the common interval $[-15, 15]\lambda_1$, without fitting the shock width. Table 11 reports raw-profile relative L^2 differences with the DSMC norm in the denominator. The comparison is descriptive: the DSMC samples are finite-time, one-seed estimates, stationarity is not certified, and higher central moments retain finite-sample bias. The DVM is a Shakhov calculation and the R26 curve is the corrected consistent Shakhov solution; equal transport coefficients do not make these collision descriptions identical. Figures 14 and 15 show the profiles. The Mach-2 figure uses raw DSMC data; the Mach-3–5 figure also shows local quadratic smoothing as a visual guide. All tabulated errors use unsmoothed profiles. The DVM curves are finite-time snapshots, not stationary references. Corrected-Shakhov R26 profiles are available and numerically accepted here at Mach 2 and 3 only; no corrected R26 result is supplied at Mach 4 or 5. Here $e_R = R_{xx,\mathrm{model}}^{\mathrm{cl}} - R_{xx,\mathrm{DSMC}}^{\mathrm{cl}}$, $e_{\Delta/3} = (\Delta_{\mathrm{model}} - \Delta_{\mathrm{DSMC}})/3$ and $e_S = S_{\mathrm{model}} - S_{\mathrm{DSMC}}$; $\langle a, b \rangle = \sum_i a_i b_i$ on the comparison interval, and c_e denotes their normalised inner product. Table 12 resolves the signed errors in $S = R_{xx}^{\mathrm{cl}} + \Delta/3$. The Mach-3 R26 row was recomputed from the accepted corrected-Shakhov profile, rather than read from a figure annotation. With the same raw-data alignment, its relative L^2 differences against 2000-ppc DSMC are 1.328%, 2.613%, 15.267%, 38.376%, 26.390% and 26.449% for $\rho, T, q_x, R_{xx}^{\mathrm{cl}}, \Delta, S$, respectively. These comparisons provide a reusable cross-method profile benchmark, not a validation of the collision-production measurement, a statistical confidence interval, or a test of the two-channel identifiability result.

Table 11. Mach-2 raw-profile relative L^2 differences against the 2000-particle-per-cell Bird DSMC result. Entries are $100\|g_{\mathrm{model}} - g_{\mathrm{DSMC}}\|_2/\|g_{\mathrm{DSMC}}\|_2$ on the common $[-15, 15]\lambda_1$ interval after density-midpoint translation only; no shock-width fit or smoothing is used in the metric. The DVM is a finite-time Shakhov snapshot and the R26 result is the corrected Shakhov profile.

Model	ρ	T	q_x	R_{xx}^{cl}	Δ	S
Shakhov DVM	0.141	0.180	2.968	16.718	25.948	8.272
Corrected Shakhov R26	0.468	0.561	6.165	22.883	25.755	11.963

Table 12. Signed-error decomposition against raw 2000-ppc DSMC profiles at Mach 2 and 3. The three error columns are $100\|e\|_2/\|S_{\mathrm{DSMC}}\|_2$ on $[-15, 15]\lambda_1$ after density-midpoint translation only; $c_e = \langle e_R, e_{\Delta/3}\rangle/(\|e_R\|_2\|e_{\Delta/3}\|_2)$. Thus $e_S = e_R + e_{\Delta/3}$ pointwise. The Mach-3 DVM row uses the saved transient snapshot at $t = 238.125$. The R26 Mach-3 row comes from the accepted corrected-Shakhov profile (SHA-256 beginning `1b870043`); the calculations use unsmoothed profiles and are descriptive finite-time comparisons.

M_1	Model	$\|e_R\|$ (%)	$\|e_{\Delta/3}\|$ (%)	$\|e_S\|$ (%)	c_e
2	Shakhov DVM	8.424	13.906	8.272	−0.836
2	Corrected Shakhov R26	11.531	13.803	11.963	−0.567
3	Shakhov DVM	8.837	13.589	9.189	−0.742
3	Corrected Shakhov R26	19.440	14.181	26.449	+0.219

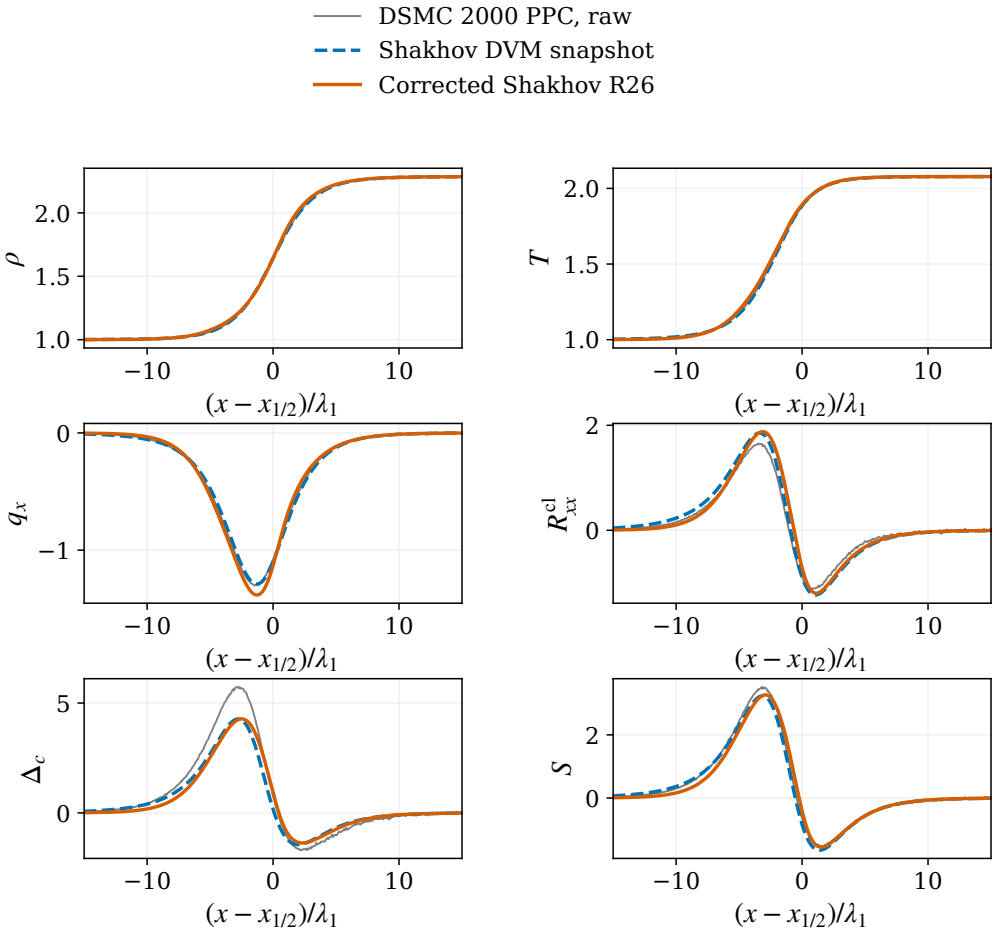


Figure 14. Mach-2 raw profiles from 2000-ppc Bird DSMC, a Shakhov DVM snapshot, and corrected-Shakhov R26. Density-midpoint translation is the only spatial alignment; the historical R26 curve preceding the symmetric trace-free (STF) corrections is excluded.

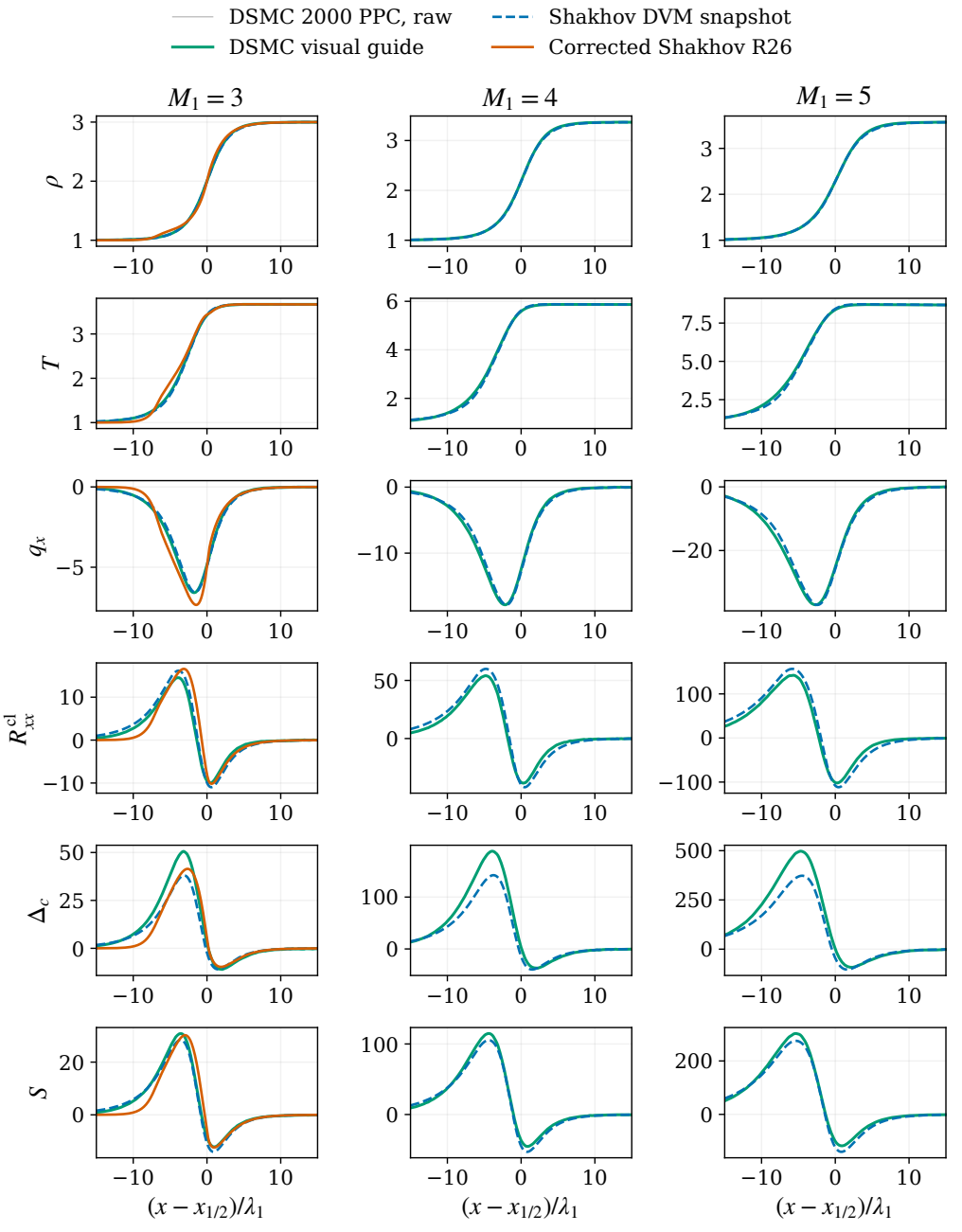


Figure 15. Mach-3–5 profile comparison using 2000-ppc Bird DSMC and Shakhov DVM snapshots, with the corrected-Shakhov R26 profile shown only where available (Mach 3). Raw DSMC data and display-only local smoothing are both shown. The DVM fields are transient snapshots; neither the DSMC stationarity nor model-specific convergence is certified.

Appendix E. Mach-8 Shakhov benchmark comparison

This appendix compares density and temperature with the S-model curves in figure 1 of Chen *et al.* (2013). The matched physical parameters are $M_1 = 8$, $Pr = 2/3$, $\omega = 0.81$ in $\mu \propto T^{\omega}$, and $\rho_1 = T_1 = R = \lambda_1 = 1$. Equation (35) of that reference gives $\mu_{\mathrm{ref}} = 1.0338373584$ and hence $\tau = \mu_{\mathrm{ref}} T^{\omega-1}/\rho$. Our calculation solves transverse-integrated Shakhov equations for $g = \int f \,\mathrm{d}v_y \mathrm{d}v_z$ and $h = \int (v_y^2 + v_z^2) f \,\mathrm{d}v_y \mathrm{d}v_z$, without clipping. The baseline uses 1200 spatial cells on $[-50, 30]\lambda_1$ and 401 longitudinal velocities on $[-40, 40]$, in double precision. Its first-order upwind discretisation differs from the reference unified gas-kinetic scheme (UGKS) discretisation with 300 spatial cells. This is a separate Mach-8 calculation from the three-dimensional-velocity BGK configurations in table 3: its collision model, relaxation law and reduced velocity representation differ, so the spatial and velocity extents are not interchangeable.

The published curves are extracted from the original PDF vector paths (133 vertices per curve), rather than supplied numerical data. Density and temperature are normalised by their Rankine–Hugoniot jumps. One translation aligns each computed density midpoint with the reference density midpoint; the same translation is used for temperature, with no fitted stretching or independent temperature shift. At the reference vertices over $-20 \le x/\lambda_1 \le 15$, the maximum absolute discrepancy, expressed as a percentage of the full jump, is reported in table 13. Figure 16 shows both profiles and differences. Finite resolution of the published curves and interpolation introduce unquantified extraction uncertainty, so sub-percent differences should not be interpreted as exact solver errors.

Table 13. Mach-8 comparison with the published Shakhov curves. Maximum discrepancies are percentages of the full density or temperature jump, after a shared density-midpoint alignment. The refined case doubles the spatial resolution and also halves the CFL time step.

Calculation	Density	Temperature
Baseline, $t = 400$	0.97	1.09
Extended time, $t = 800$	0.97	1.09
Spatial refinement, $N_x = 2400$, $t = 400$	0.78	0.99

The close macroscopic agreement is a bounded benchmark check, not validation of a nonnegative kinetic distribution or of the fourth-order channels. Here Y denotes any field in the five-field list below. In all cases the prescribed stationarity criterion remains unmet: every 2000 steps, the maximum profile change over $(\rho, u_x, T, q_x, \sigma_{xx})$, divided fieldwise by $\max(1, \max_x |Y|)$, must be below 10^{-7} for five consecutive checks on the fixed spatial grid. The integrations terminate at their prescribed physical times. A small translation of the shock contributes to this unaligned criterion even when aligned profiles are close.

The six sensitivity calculations test extended integration time, finer spatial resolution, a smaller time step, finer velocity resolution, a larger velocity domain and a larger spatial domain. The table and figure report comparisons for the baseline, extended-time and spatially refined calculations. Across the six tests, the maximum local negative-g mass fraction remains approximately 2.13–2.15%, and the corresponding negative-h energy fraction is 5.15–5.22%. Small aligned macroscopic sensitivities do not remove this positivity limitation. The reduced distributions also do not determine all three-dimensional fourth-order moments. This appendix therefore supports the reported density/temperature comparison only; it does not independently validate R_{xx}^{cl}, Δ or S, or replace the constant-relaxation BGK high-Mach results in §5.4.

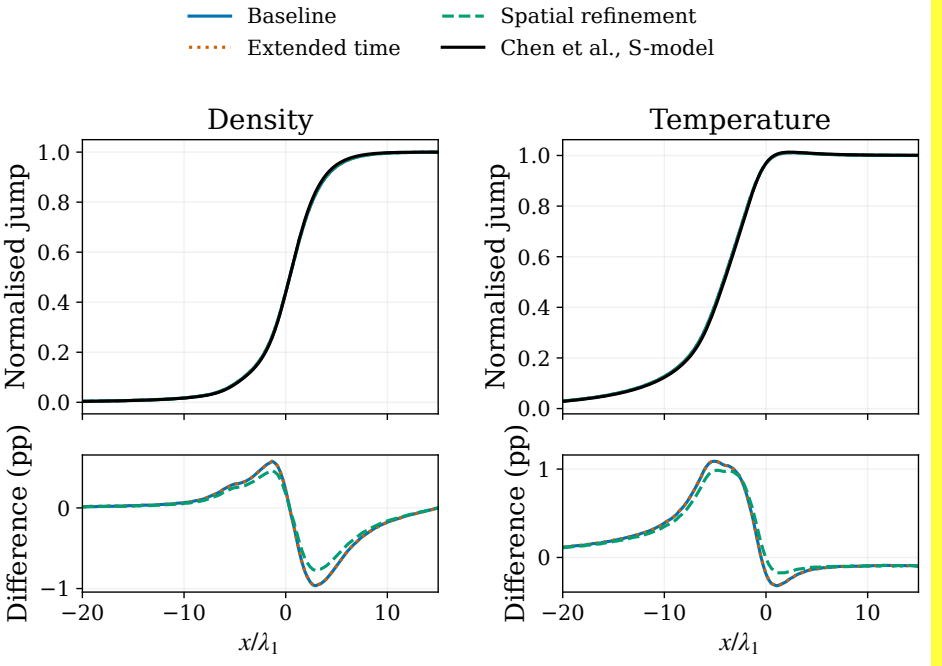


Figure 16. Matched-parameter Mach-8 Shakhov benchmark comparison. Black curves are the vector-extracted S-model results from figure 1 of Chen *et al.* (2013); coloured curves are the present baseline, time-extended and spatially refined calculations. Lower panels show discrepancies in percentage points of the normalised jump. Density and temperature use the same spatial translation for each calculation.

153 ## Appendix F. Normal-shock symmetry and supporting moment identities

Symmetry assumptions. The planar normal-shock problem has no transverse flow, forcing or preferred transverse direction. Its Maxwellian reservoirs and isotropic collision law are invariant under transverse reflections and rotations. We consider the solution branch that preserves these symmetries, as does evolution from symmetry-preserving initial data. This specifies the branch; it is not a uniqueness theorem for every steady solution of the kinetic boundary-value problem. At each fixed x, write $c_y = v_y$, $c_z = v_z$ and $c_x = v_x - u_x(x)$. The following three symmetries suffice for the moment reduction:

$$\begin{aligned} f(c_x, c_y, c_z) &= f(c_x, -c_y, c_z) = f(c_x, c_y, -c_z), \\ f(c_x, c_y, c_z) &= f(c_x, c_z, c_y). \end{aligned} \tag{F 1}$$

The local equilibrium reference has the same symmetries. No reflection symmetry in c_x is assumed: the streamwise heat flux may therefore be non-zero.

Componentwise proof. Extend (2.2) and (2.3) to the full tensors, with δ_{ij} the Kronecker delta:

$$\begin{aligned} C_{ij}(c) &= c_i c_j - \tfrac{1}{3}|c|^2 \delta_{ij}, \qquad \sigma_{ij} = \int C_{ij} f \, \mathrm{d}v, \\ R^{\mathrm{cl}}_{ij} &= \int |c|^2 C_{ij} (f - M_d) \, \mathrm{d}v - 7T\sigma_{ij}. \end{aligned} \tag{F 2}$$

For $i \neq j$, $C_{ij} = c_i c_j$. Changing $c_y \to -c_y$ reverses the sign of the xy and yz integrands without changing f, M_d or the integration measure; changing $c_z \to -c_z$ does the same for xz. Each such integral equals its negative and is zero. This proves that the off-diagonal components of both σ and R^{cl} vanish. Next, changing variables $c_y \leftrightarrow c_z$ interchanges the yy and zz integrands while leaving the distributions and measure unchanged, giving equality of their integrals. Finally, the pointwise identity

$$\sum_{i=x,y,z} C_{ii} = |c|^2 - |c|^2 = 0$$

implies $\sum_i \sigma_{ii} = 0$ and $\sum_i R^{\mathrm{cl}}_{ii} = 0$, including the stress subtraction. Combining these facts yields

$$R^{\mathrm{cl}}_{yy} = R^{\mathrm{cl}}_{zz} = -\tfrac{1}{2} R^{\mathrm{cl}}_{xx}, \qquad \sigma_{yy} = \sigma_{zz} = -\tfrac{1}{2}\sigma_{xx}, \qquad R^{\mathrm{cl}}_{ij} = \sigma_{ij} = 0 \quad (i \neq j). \tag{F 3}$$

Consequently $\mathrm{tr}(\sigma^2) = \sigma_{xx}^2 + 2(\sigma_{xx}/2)^2 = 3\sigma_{xx}^2/2$, as used in (5.11). These restrictions leave Δ independent: it is a scalar integral, not the trace of the already trace-free tensor R^{cl}. For example, on full velocity space with continuum-reference moments, a mixture of isotropic Maxwellians with different temperatures and the same mean velocity has $R^{\mathrm{cl}}_{ij} = \sigma_{ij} = 0$. For positive mixture fractions a_ℓ summing to one and component temperatures T_ℓ, the mixture temperature is $T = \sum_\ell a_\ell T_\ell$ and $\Delta_c = 15\rho[\sum_\ell a_\ell T_\ell^2 - T^2] > 0$ when the temperatures differ. This illustrates independence at the level of moment definitions, not an additional shock solution.

Discrete velocities. The same change-of-variables proof becomes a node-pairing proof when the transverse node sets and weights are invariant under sign changes and interchange, and the sampled distribution and fitted M_d preserve those symmetries. Equal y and z grids symmetric about zero admit these pairings; invariance under arbitrary rotations is unnecessary. The trace identity holds nodewise even on a finite grid. The other equalities require the stated symmetries and are subject to numerical symmetry errors; finite-particle DSMC samples need not satisfy them exactly. In particular, symmetry enforces equality of the two transverse stresses, not vanishing diagonal stress of a finite-grid Maxwellian. It does not remove the equilibrium-quadrature defects reported in table 7.

Why the coefficient 7? It is a continuum R26 moment convention. Let M be the Maxwellian with covariance TI and consider only its standard stress-polynomial contribution

$$f_\sigma = M\frac{\sigma_{kl}c_k c_l}{2\rho T^2}, \qquad \sigma_{kk} = 0,$$

where repeated indices are summed. Gaussian contraction gives

$$\int |c|^2 c_i c_j c_k c_l M \,\mathrm{d}v = 7\rho T^3(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}). \tag{F 4}$$

To verify the factor, write $|c|^2 = \sum_a c_a^2$: in the Gaussian pairings of $c_a c_a c_i c_j c_k c_l$, pairing the two a indices together contributes the dimension 3, while the remaining pairings contribute 4, giving $3+4=7$ for each displayed Kronecker product. Contracting with $\sigma_{kl}/(2\rho T^2)$ removes the first product and gives $7T\sigma_{ij}$. The scalar trace contribution vanishes under the same contraction, so

$$\int |c|^2 C_{ij} f_\sigma \,\mathrm{d}v = 7T\sigma_{ij}.$$

Subtracting this contribution defines the fourth-order departure used in (F 2); it does not assume the actual shock distribution equals a truncated Grad expansion. On a finite grid the Gaussian contraction need not be exact. The discrete-reference definition retains the coefficient 7 as the chosen moment convention, with the separate M_d subtraction explicitly specified in (2.3).

Appendix G. Independently evolved ES-BGK shocks at Mach 2 and 3

The source evaluations in §5 are supplemented by independent ES-BGK shock evolutions. These calculations use the anisotropic Gaussian target in (5.7), rather than substituting its source into a saved BGK solution. The upstream state is $\rho_1 = T_1 = 1$, $R = 1$, $\gamma = 5/3$ and $u_1 = M_1\sqrt{5/3}$; incoming populations at the two boundaries are equilibrium Rankine–Hugoniot reservoirs. With $Pr = 2/3$ and $\alpha_{\mathrm{ES}} = 1 - 1/Pr = -1/2$, viscosity matching to the Shakhov law $\mu = T$ requires

$$\tau_{\mathrm{ES}} = (1-\alpha_{\mathrm{ES}})\frac{\mu}{p} = \frac{3}{2\rho}, \qquad p = \rho T. \tag{G 1}$$

Using $\tau_{\mathrm{ES}} = \mu/p$ would give a different viscosity. Positions use the shared Shakhov upstream relaxation-length reference, $\lambda_1 = \sqrt{RT_1}\,\tau_{\mathrm{Sh},1} = 1$ in computational units; the different ES-BGK relaxation time is not used to rescale the plotted coordinates.

The implemented kinetic solver evolves three reduced distributions, $g = \int f\,\mathrm{d}v_y\mathrm{d}v_z$, $h = \int (c_y^2 + c_z^2) f\,\mathrm{d}v_y\mathrm{d}v_z$ and $k = \int (c_y^2 + c_z^2)^2 f\,\mathrm{d}v_y\mathrm{d}v_z$. Normal-shock symmetry gives zero transverse mean velocity. The transverse Gaussian target integrals are evaluated analytically: $h_G = 2T_\perp g_G$ and $k_G = 8T_\perp^2 g_G$, with $T_\parallel = T + \alpha_{\mathrm{ES}}\sigma_{xx}/\rho$ and $T_\perp = T - \alpha_{\mathrm{ES}}\sigma_{xx}/(2\rho)$. This reduction retains the axial and contracted fourth moments used here. Spatial transport uses conservative first-order upwind finite volumes with one shared interface flux; collision uses a self-consistent backward-Euler update with implicit stress relaxation. The axial velocity interval is $[-40, 40]$, with trapezoidal quadrature. There is no clipping, silent normalisation or periodic recentering in these ES-BGK calculations.

The Mach-2 and Mach-3 domains are $[-30, 30]$ and $[-45, 45]$, respectively. Baseline cells have $\Delta x = 0.1875$, axial velocity spacing $\Delta v = 0.2$ and $\Delta t = 0.00375$ ($C_{\mathrm{CFL}} = 0.8$). The spatial-refinement cases halve both Δx and the CFL-selected time step; velocity refinement then halves Δv at that finer spatial grid. Thus the baseline-to-spatial-refinement comparison changes space and time together. The six completed cases in table 14 terminate at $t \simeq 400$. Separate half-time-step checks and subsequent continuations are excluded from this frozen result set because they were not completed at the time of this deposit.

Stationarity requires five consecutive comparisons approximately two physical-time units apart, with the largest scaled change in each of $\rho, u_x, T, q_x, \sigma_{xx}, R_{xx}^{\mathrm{cl}}, \Delta, S$ below 10^{-7} and the scaled moment rate from a full numerical step below 10^{-7}. Each family is scaled by its maximum absolute value, with a 10^{-8} denominator floor. An additional numerical step at the final saved state measures the reported post-state RHS. None of these six cases passes this criterion: their final rates remain between 5.5×10^{-5} and 6.2×10^{-5}. Positive density, temperature, target covariance and reduced populations are retained, but successful execution and positivity do not certify a stationary shock or global stability.

Table 14. Completed finite-time ES-BGK cases at $Pr = 2/3$, $t \simeq 400$. B denotes baseline, X coupled spatial/time refinement, and V subsequent axial-velocity refinement. N_x counts cells and N_v axial velocity nodes. The three errors are percentages relative to direct velocity moments on the common mask $|S_{\mathrm{direct}}| > 0.02 \max|S_{\mathrm{direct}}|$. They measure steady-form budget reconstruction of a nonstationary snapshot, not ES-BGK model error. RHS is the final scaled full-step moment rate.

Case	N_x	N_v	Δt	ε_S	ε_Δ	ε_R	RHS
M2 B	320	401	0.003750	13.70	16.62	26.62	6.15×10^{-5}
M2 X	640	401	0.001875	6.80	8.46	13.24	6.15×10^{-5}
M2 V	640	801	0.001875	6.82	8.47	13.27	6.15×10^{-5}
M3 B	480	401	0.003750	8.35	12.10	17.19	5.53×10^{-5}
M3 X	960	401	0.001875	4.22	6.15	8.71	5.57×10^{-5}
M3 V	960	801	0.001875	4.22	6.16	8.71	5.57×10^{-5}

Direct R_{xx}^{cl} and Δ use the continuum-reference definitions, and $S = R_{xx}^{\mathrm{cl}} + \Delta/3$. The heat reconstruction integrates $Q_q - (\partial_x u_x)L_q$ using the actual upstream F_q anchor and trapezoidal spatial integration. The scalar reconstruction uses the actual finite-velocity-grid ES target excess, $\Delta_G = \sum_j w_j(c_{x,j}^4 g_{G,j} + 2c_{x,j}^2 h_{G,j} + k_{G,j}) - 15\rho T^2$, and the derivative of J_Δ plus $4(\partial_x u_x)K_\Delta$; the recovered tensorial moment is $R_{\mathrm{bud}}^{\mathrm{cl}} = S_{\mathrm{bud}} - \Delta_{\mathrm{bud}}/3$. Here j indexes axial velocity nodes, w_j are their quadrature weights, $c_{x,j} = v_j - u_x$,

and $g_{G,j}, h_{G,j}, k_{G,j}$ are the three reduced ES target distributions at a node. The fluxes are the reduced-distribution equivalents of (5.2); derivatives use centred differences in the interior and second-order one-sided boundary formulas. The same gas mean is used for the target and distribution moments. On the completed states, the finite-grid target heat flux is below 1.5×10^{-14} and the discrepancy from $\Delta_G = 3\alpha_{\mathrm{ES}}^2 \sigma_{xx}^2/\rho$ is below 9×10^{-12}.

The errors are $100\|g_{\mathrm{bud}} - g_{\mathrm{direct}}\|_{2,\mathscr{A}} / \|g_{\mathrm{direct}}\|_{2,\mathscr{A}}$ for each $g = S, \Delta, R_{xx}^{\mathrm{cl}}$, on the common mask $\mathscr{A}$ defined in the table. No separate curve alignment or plateau subtraction is applied to these budget checks. Coupled spatial/time refinement approximately halves the reconstruction discrepancies, while subsequent velocity refinement changes them little. Because the reconstructions omit time derivatives, their mismatch includes temporal evolution as well as discretisation and derivative/integration errors; it does not independently validate the steady budget or quantify collision-model error.

Figures 17 and 18 compare the six displayed fields with the archived DSMC, Shakhov DVM and corrected Shakhov R26 profiles used in appendix D. Each curve is shifted once to its density midpoint. Density and temperature are close across the models, whereas the fourth moments show more visible differences. These are comparisons between different collision models and numerical grids, with finite-time and sampling limitations; they are not matched-grid convergence or exact-error studies. The common observation channel does not require the different models to produce identical R_{xx}^{cl}, Δ or S profiles.

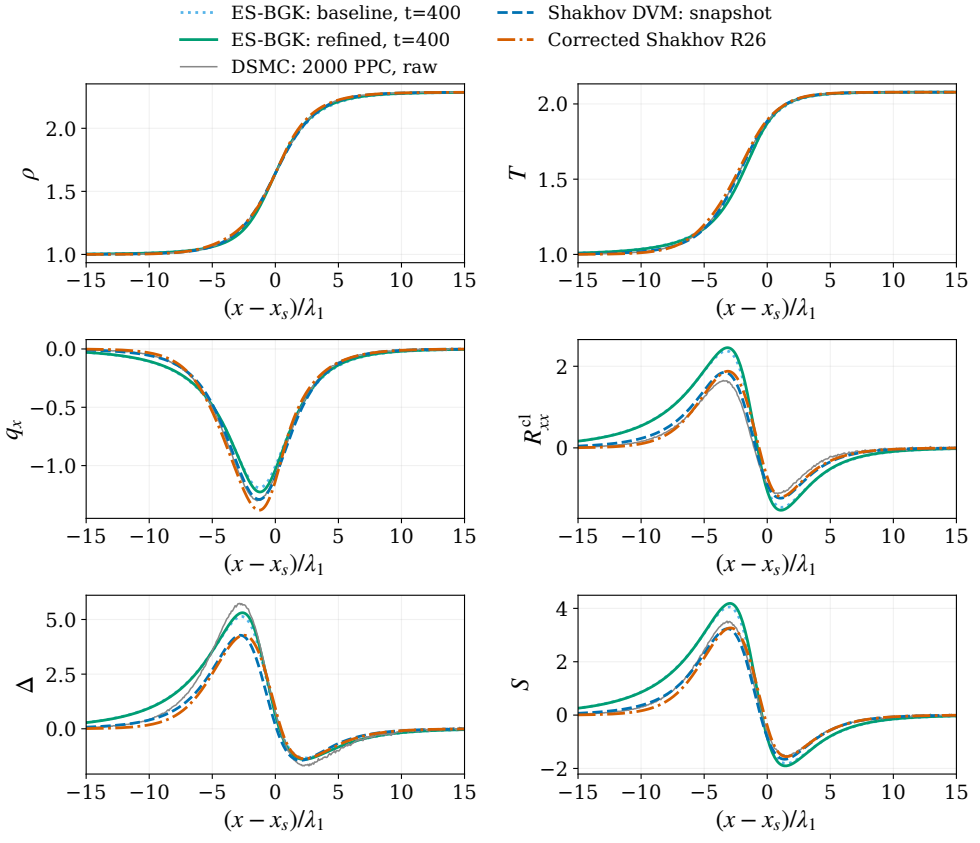


Figure 17. Mach-2 finite-time ES-BGK profiles ($Pr = 2/3$, $\mu = T$, $t \simeq 400$) compared with archived 2000-PPC DSMC, Shakhov DVM and corrected Shakhov R26 profiles. The ES baseline uses $N_x = 320$, $N_v = 401$; the refined curve uses $N_x = 640$, $N_v = 801$. Only density-midpoint translations are applied; raw DSMC profiles are not smoothed. No plotted ES-BGK case has passed the stationarity criterion. The different collision models and grids prevent interpreting curve differences as numerical error alone.

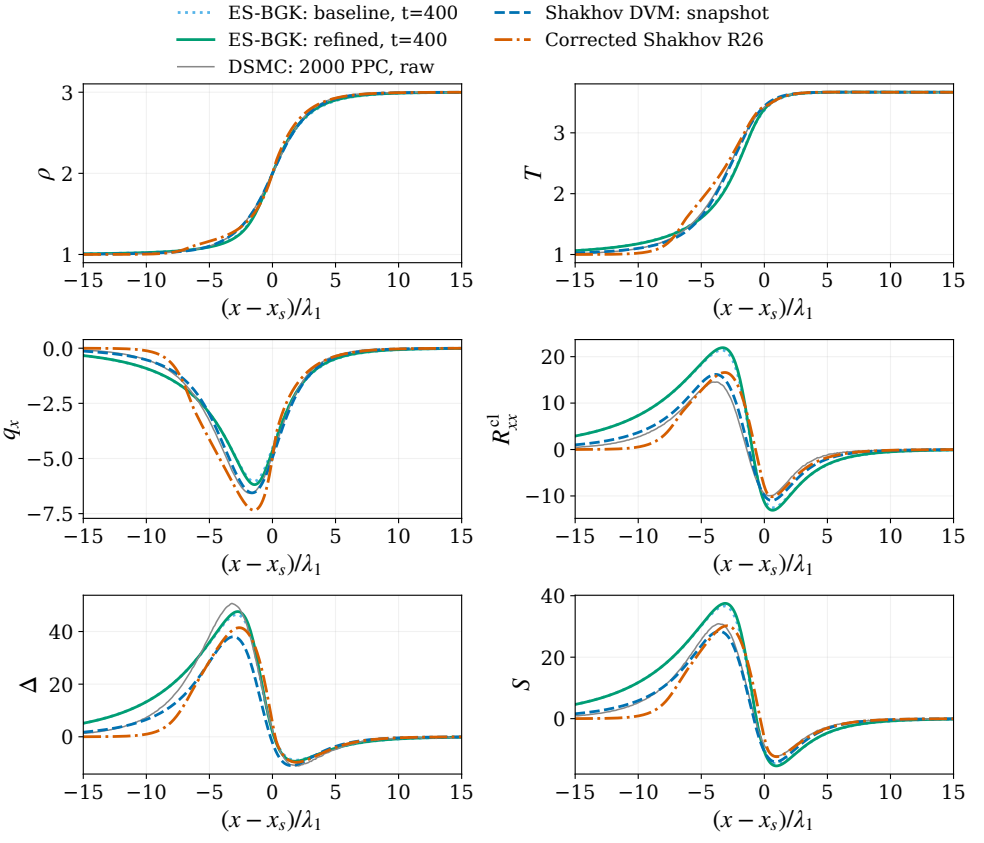


Figure 18. Mach-3 counterpart of figure 17. The ES-BGK baseline uses $N_x = 480$, $N_v = 401$; the refined curve uses $N_x = 960$, $N_v = 801$. Model transport parameters, finite-time qualification and plotting conventions are the same as in figure 17. The archived corrected R26 profile is a separate finite-residual-qualified Shakhov calculation.

References

Alsmeyer, H. 1976 Density profiles in argon and nitrogen shock waves measured by the absorption of an electron beam. *Journal of Fluid Mechanics* **74** (3), 497–513.

Andries, Pierre, Le Tallec, Patrick, Perlat, Jean-Philippe & Perthame, Benoit 2000 The gaussian-BGK model of boltzmann equation with small prandtl number. *European Journal of Mechanics B/Fluids* **19** (6), 813–830.

Bhatnagar, P. L., Gross, E. P. & Krook, M. 1954 A model for collision processes in gases. i. small amplitude processes in charged and neutral one-component systems. *Physical Review* **94** (3), 511–525.

Bird, Graeme A. 1994 *Molecular Gas Dynamics and the Direct Simulation of Gas Flows*. Oxford: Clarendon Press.

Cercignani, Carlo 1988 *The Boltzmann Equation and Its Applications*. Springer.

Chen, Songze, Xu, Kun & Cai, Qingdong 2013 A comparison and unification of ellipsoidal statistical and shakhov BGK models. ArXiv:1304.0865.

Dimarco, Giacomo & Pareschi, Lorenzo 2014 Numerical methods for kinetic equations. *Acta Numerica* **23**, 369–520.

Fei, Fei, Liu, Haihong, Liu, Zhaohui & Zhang, Jun 2020 A benchmark study of kinetic models for shock waves. *AIAA Journal* **58** (6), 2596–2608.

Filbet, Francis & Jin, Shi 2010 A class of asymptotic-preserving schemes for kinetic equations and related problems with stiff sources. *Journal of Computational Physics* **229** (20), 7625–7648.

Grad, Harold 1949 On the kinetic theory of rarefied gases. *Communications on Pure and Applied Mathematics* **2** (4), 331–407.

Gu, Xiao-Jun & Emerson, David R. 2009 A high-order moment approach for capturing non-equilibrium phenomena in the transition regime. *Journal of Fluid Mechanics* **636**, 177–216.

Han, Jiequn, Ma, Chao, Ma, Zheng & E, Weinan 2019 Uniformly accurate machine learning-based hydrodynamic models for kinetic equations. *Proceedings of the National Academy of Sciences* **116** (44), 21983–21991.

Holway, Lowell H. 1966 New statistical models for kinetic theory: methods of construction. *Physics of Fluids* **9** (9), 1658–1673.

Huang, Juntao, Cheng, Yingda, Christlieb, Andrew J., Roberts, Luke F. & Yong, Wen-An 2023 Machine learning moment closure models for the radiative transfer equation II: enforcing global hyperbolicity in gradient-based closures. *Multiscale Modeling & Simulation* **21** (2), 489–512.

Ivanov, M. S. & Gimelshein, S. F. 1998 Computational hypersonic rarefied flows. *Annual Review of Fluid Mechanics* **30**, 469–505.

Karniadakis, George Em, Kevrekidis, Ioannis G., Lu, Lu, Perdikaris, Paris, Wang, Sifan & Yang, Liu 2021 Physics-informed machine learning. *Nature Reviews Physics* **3**, 422–440.

Levermore, C. David 1996 Moment closure hierarchies for kinetic theories. *Journal of Statistical Physics* **83**, 1021–1065.

Lou, Qin, Meng, Xuhui & Karniadakis, George Em 2021 Physics-informed neural networks for solving forward and inverse flow problems via the Boltzmann–BGK formulation. *Journal of Computational Physics* **447**, 110676.

Mathiaud, Julien & Mieussens, Luc 2016 A fokker–planck model of the boltzmann equation with correct prandtl number. *Journal of Statistical Physics* **162**, 397–414.

Mieussens, Luc 2000 Discrete velocity model and implicit scheme for the BGK equation of rarefied gas dynamics. *Mathematical Models and Methods in Applied Sciences* **10** (8), 1121–1149.

Mott-Smith, Harold M. 1951 The solution of the Boltzmann equation for a shock wave. *Physical Review* **82** (6), 885–892.

Oh, Jaemin, Cho, Seung Yeon, Yun, Seok-Bae, Park, Eunbyung & Hong, Youngjoon 2025 Separable physics-informed neural networks for solving the BGK model of the Boltzmann equation. *SIAM Journal on Scientific Computing* **47** (2), C451–C474.

Raissi, Maziar, Perdikaris, Paris & Karniadakis, George Em 2019 Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* **378**, 686–707.

Roohi, Ehsan 2026*a* Closure-level observability in neural kinetic shock solvers. ArXiv:2605.29211, arXiv: 2605.29211.

Roohi, Ehsan 2026*b* DVM-JFM: archived solvers, compact profiles and provenance for closure-channel recovery in kinetic normal shocks. https://github.com/Ehsan-Roohi/DVM-JFM, scientific deposit, commit 4c3d3118cd77a780049f04e80ef5f05f216aa2e4; accompanying manuscript revision, tag reviewer-revision-20261010, 10 October 2026. Raw-distribution exclusions and reproducibility limitations are documented in the repository.

SHAKHOV, E. M. 1968 Generalization of the Krook kinetic relaxation equation. *Fluid Dynamics* **3** (5), 95–96.
SONE, YOSHIO 2007 *Molecular Gas Dynamics: Theory, Techniques, and Applications*. Birkhäuser.
STRUCHTRUP, HENNING 2005 *Macroscopic Transport Equations for Rarefied Gas Flows*. Springer.
STRUCHTRUP, HENNING & TORRILHON, MANUEL 2003 Regularization of Grad's 13 moment equations: Derivation and linear analysis. *Physics of Fluids* **15** (9), 2668–2680.
STRUCHTRUP, HENNING & TORRILHON, MANUEL 2007 H theorem, regularization, and boundary conditions for linearized 13 moment equations. *Physical Review Letters* **99**, 014502.
TORRILHON, MANUEL 2016 Modeling nonequilibrium gas flow based on moment equations. *Annual Review of Fluid Mechanics* **48**, 429–458.
TORRILHON, MANUEL & STRUCHTRUP, HENNING 2004 Regularized 13-moment equations: shock structure calculations and comparison to Burnett models. *Journal of Fluid Mechanics* **513**, 171–198.